1. Excellence

The **overall project objectives** are; ***(1)*** To develop and demonstrate robust, scalable high yield (>15%) innovative manufacturing processes for critical steps in production of opto-electronic devices including OLEDs, OPVs and CIGs solar cells in S2S and R2R environment. **(2)** A complete inline hybrid system of inspection, quality control, functional testing and measurements using advance techniques and sensors will be optimised in the project for efficient manufacturing of opto-electronics parts with minimum defects resulting in reduction in production cost and time. ***(3)*** Recycling and re-use approaches and strategies will be developed in the project allowing resources efficiency and reduction of production of high value product wastes (generating >10% of production cost through recycled components). ***(4)*** Automation and advance processing software will be developed for overall control and monitoring of manufacturing of OLEDs, OPVs and CIGs solar cells in R2R and S2S manufacturing process allowing a time of reduction in production processes reconfiguration by more than 15%. The proposed activities will be scaled up, tested in 3 pilot lines and implemented in 4 production lines for validation.

Background: Opto-electronic devices are opening exciting new applications. With new display options using pliable substrates such as plastic and flexible glass, organic light-emitting diodes (OLEDs) manufacturers are bringing a wide range of new applications in lighting (e.g. energy efficient lighting, eco-friendly lighting, lighting panels) and displays (e.g. small displays for mobile applications, TVs and micro-displays). Similarly, with the emergence of thin-film technologies in the solar cells market, new applications ranging from installations on curved surfaces to building-integrated photovoltaics became possible. However, to meet the industry requirements required for mass production, including price points, manufacturing volumes and efficiency, many challenges still need to be addressed.

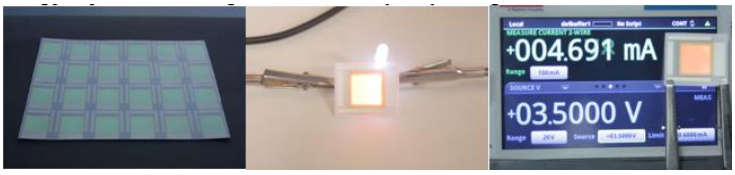


Figure 1 OLED products of INURU

Industrial need and challenges: The challenges associated to **OLEDs** include manufacturing cost reduction, equipment depreciation, barrier films performance issues, blue emitter issues in devices and loss due to waste through inefficient deposition and low yields etc. **Organic photovoltaics (OPVs)** can be fabricated as thin flexible sheets using conventional printing techniques and have the potential of saving significant materials and costs as well as reducing environmental impact. Due to their evolving power conversion efficiencies and integrability of OPVs, it has the potential to enable new market applications. But there are some challenges and needs associated to cost effective production of OPVs including manufacturing processes optimisation, metrology and inspection allowing defect free OPVs manufacturing, automation of production processes, efficient coatings and barriers performance, production scale up issues to achieve cost effective high yield production. **Thin-film photovoltaic (PV) modules of CIGS** have the potential to reach cost-effective PV-generated electricity. These technologies have transitioned from the laboratory to the market place. Pilot production and first-time manufacturing are ramping up to higher capacity and enjoying a flood of venture-capital funding. Solar modules from **Solibro (***End user in OLEDSOLAR consortium***)** has broken efficiency records – double glass CIGS solar modules at 18.72 percent and sputtering CIGS solar modules on flexible substrate at 17.88 percent[1]. Manufacturing-line yield continues to improve and is surpassing 85%[2]. Long-term stability has been demonstrated for both technologies; however, some failures in the field have also been observed, emphasizing the critical need for understanding metrology and packaging options. Mostly, thin-film technologies have a common device/module structure: substrate, base electrode, absorber, junction layer, top electrode, patterning steps and encapsulation. The long-term potential of solar cell technologies requires addressing the need and challenges e.g. in-situ process control and diagnostics, thinner absorber, understanding degradation mechanisms, protection from water vapor, and innovation in high-speed processing and module design to achieve targeted cost-effective module performance, and in-field durability. To address such challenges **solutions that have shown potential[3] are;** (1) Low cost and fast-

Figure 2 OPV products and applications cases of ASPF

[1] http://clikhe.re/mbtqe#https://inhabitat.com/hanergys-thin-solar-panels-break-multiple-world-records-for-efficiency/
[2] http://ieeexplore.ieee.org/document/4059628/?reload=true
[3] http://ieeexplore.ieee.org/document/4059628/?reload=true

deposition processes for CIGS, (2) improved metrology and quality control measures, (3) advanced NDT and process control and understanding of material degradation and aging, etc.

Figure 3 Solibro's CIGs solar cell

Thus, the **biggest challenges in the thin-film solar cells market (CIGS) and the organic electronics industry, organic solar cells and organic LEDs,** are **how to scale-up from laboratory to mass production, how to select the most efficient manufacturing processes, and how to use inspection, control and measurement techniques to improve yield, quality and time-to-market.** The transition from small production lines to mass production requires adjusting and optimizing many different parameters and processes along the value chain, both at material (e.g. deposition level, substrates, ink and solvent systems), manufacturing (e.g. roll-to-roll manufacturing) and inspection & quality control level. Each parameter has its influence on the performance, lifetime and efficiency of the final electro-optical component. For roll-to-roll production of these components, the correlation between all parameters and the final device performance needs to be determined as early as possible. For instance, the performance of **organic solar cells and organic LEDs** is highly dependent on the thicknesses of the constituent layer materials composing the device. The lack of accurate information about the thickness of the thin films would prevent the detection of components that not perform well against design specifications. Without real-time thin film thickness measurements, or continuous yield monitoring directly integrated in the production line, the capacity of optimizing the production process becomes very limited and costly.

Market Opportunity: OLED market is expected to be valued at 40 Billion euros by 2023, growing at a CAGR of 15.2% between 2017 and 2023[4]. The key factors driving the growth of the market include the rapid adoption of OLED displays in smartphones and growing investments in technology and manufacturing facilities. Thin film solar cells market size globally was over €6.5 billion euros in 2015 with over 16% growth forecast from 2016 to 2024[5]. **CIGS market** share is **expected to exceed 6 billion euros by 2024** owing to its high absorption coefficient features.[6] The photovoltaics market is expected to grow from €71 billion in 2013 to €278 billion by 2020, at a CAGR of 18.30% between 2014 and 2020[7]. Where, the **Organic Photovoltaics (OPV) market is** expected to reach **€350 million by 2021[8]**. Thus, it presents the significant opportunities for **OLEDSOLAR** proposed manufacturing, metrology, quality assurance, automation and recycling solutions for high performance and cost-effective production of high value devices.

1.1 Objectives

The specific project objectives with relevant KPIs[9] and are listed below:

Specific Technical Objectives:

Manufacturing processes:

1. To develop an adaptive material handling system for production lines handling OLEDs, OPVs, CIGS and film manufacturing, *(KPI: 70% process improvement in pick & place, lamination tool and pressure transfer nozzle) (by month 8)*
2. To optimise R2R and S2S gravure, screen, flexo-printing and R2R slot die coating processes for OLEDs and solar cells, *(KPI: 100% automated allowing freedom of design and customisation of OPVs devices) (by month 16)*
3. To scale up spatial atomic layer deposition process (currently at TRL5) for R2R and S2S production of OLEDs and solar cells for deposition of layers and encapsulation ensuring complete protection of cells against humidity, *(KPI: High deposition rates in nm/s,>98% yield, R2R and S2S capability, and no vacuum requirement etc) (by month 28)*
4. To develop a low temperature innovative Plasma Enhanced Spatial Atomic Layer Deposition process (currently at TRL4) for moisture sensitive devices on flexible web for Roll to Roll (R2R) manufacturing of OLEDs and solar cells, *(KPI: low temperature deposition on moisture sensitive devices on flexible web)(by month 28)*
5. To optimise and test inkjet-based deposition techniques for encapsulation of OLEDs and solar cells, *(KPI: To achieve high quality encapsulation of devices) (by month 21)*
6. To optimise, automate and test laser, lamination and packaging processes for devices (OLEDs and solar cells) fabrication, *(KPI: High quality cost effective packaging) (by month 21)*
7. To optimise manufacturing processes of barrier films for encapsulation of opto-electronic devices, *(KPI: 100% defect free development, and handling of barrier films in varying conditions, cost <8€/m^2) (by month 28)*

[4] https://www.marketsandmarkets.com/PressReleases/global-OLED-market.asp
[5] https://www.gminsights.com/industry-analysis/thin-film-solar-cells-market
[6] https://www.gminsights.com/industry-analysis/thin-film-solar-cells-market
[7] https://www.marketsandmarkets.com/PressReleases/pv-market.asp
[8] http://www.marketresearchstore.com/report/global-organic-photovoltaics-opv-market-rport-2017-154896
[9] Relevant indicators and metrics for project solution targeted performance and impact implementation

In-Line inspection, metrology and quality control:

8. To optimise and implement in-line optical inspection systems based on spectroscopic imaging ellipsometry and optical coherence tomography for nano-scale monitoring (up to 200nm dry) and testing of OLEDs and solar cells components manufacturing in R2R and S2S production environment, *(KPI: to monitor up to 200nm scale resolution, up to >300mm web width with 99% probability of defect detection)(by month 20)*
9. To optimise and implement in-line NDT testing including digital shearography and thermography (dark and illuminated lock-in thermography) for zero defects quality assurance at device level for OLEDs and solar cells manufacturing, *(KPI: 100% surface and inner layer defects detection)(by month 19)*
10. To develop and implement automated in-line Raman spectroscopy and photo-luminescence testing process for optical properties validation and measurements of OLEDs and solar cells, *(KPI: 100% correlate measurements and inspection data including light performance and thickness measurement)(by month 17)*
11. To optimise and deploy testing and validation processes for qualification of thin film barrier coatings for flexible opto-electronic devices including water vapour transmission rate analysis, hyperspectral imaging (HSI) and functional measurements for validating barrier film performance, *(KPI: to achieve analysis of gas barrier films down to 10^{-6} g/m²×d) (by month 17)*
12. To optimise and implement functional testing and measurement systems for different parameters in-situ production of OLEDs and solar cells including aging tests. *(KPI: High speed and 98% accuracy)(by month 28)*
13. To optimise and scale up large area printing quality measurement and machine vision based micro-meter level register accuracy measurement tool for visible and highly transparent layers in opto-electronic devices manufacturing, *(KPI: 100% automated process and 99% accuracy)(by month 23)*
14. To develop and implement machine learning based inspection data processing and automated defect detection software inline manufacturing process of OLEDs and solar cells, (*KPI: Improving defect detection rate up to 98%) (by month 28)*

Process Automation, monitoring and control:

15. To implement sensor-based network for automation and monitoring of production processes of OLEDs and solar cells, *(KPI: To achieve >90% process automation steps) (by month 20)*
16. To implement intelligent AI based decision management and control software for automation and control of production processes of OLEDs and solar cells. *(KPI: >95% automated decision management system, Process reconfiguration time reduction due to change of component requirement by 15%) (by month 23)*

Recycling and Re-use:

17. To evaluate and implement scalable recycling solution including pre-treatment strategy, sorting and re-use strategies including LCA and LCC analysis of production devices*(KPI: >10% of production cost to be covered through recycled parts and materials) (by month 26)*
18. To develop repair and re-use strategies for high value OLEDs and solar cells (for rejected devices with minor defects) to enhance production yield and reduce waste, emission and production cost losses,(KPI: *98% recycling or re-use of rejected parts) (by month 26)*

Demonstrations: *(KPIs: >15% Yield Improvement,>15% process reconfiguration time reduction,>10% production cost covered through recycling of tangible parts)*

19. R2R production pilot line demonstration of OLEDSOLAR systems and processes, (at VTT's pilot line) *(by month 36)*
20. Pilot production line demonstration of process automation and control, (at VTT's, Coatema' s and IRIS's pilot line) *(by month 36)*
21. R2R and S2S production pilot line testing and demonstration of manufacturing processes (at TNO and SMIT TS pilot lines) *(by month 36)*
22. Industrial testing at pilot line of CIGS PVs, (at TNO/FhG/SAL) *(by month 36)*
23. Industrial testing at production line of OPVs, (at ASPF, end user) *(by month 36)*
24. Industrial testing at production line of OLEDs, (at INURU, end user) *(by month 36)*

Commercial Objectives:

25. To develop Exploitation and business plan for successful implementation and market transfer of the innovative OLEDSOLAR system by month 36.
26. Dissemination of the project outcome to at least 6 international events before month 36. And dissemination/awareness campaign and knowledge transfer events (at least 6 events) for industrial stakeholders from second year of project till end.
27. Compliance and Implementation of governing production and factories/workers safety regulations and standards require for commercialization.*(by month 36)*

Innovative approaches proposed for achieving the above stated objectives:

- Automated material handling system for substrate handling, devices fabrication and packaging automation resulting in production labour cost reduction and increasing production time efficiency,
- Innovative, automated printing and laser processes for material patterning in solar cells.
- Innovative high speed spatial and plasma enhanced atomic layer deposition techniques and water transmission measurement techniques enabling efficient automated R2R production of OLEDs and solar cells allowing high quality, customisation and production process adaptability as per customer needs.
- Cost effective 100% in-line inspection of optical components during production simultaneously at production speed and frequency through innovative optimised set of inspection and measurement techniques with advance machine learning data processing and defects detection techniques.
- Innovative process automation sensor system combined with intelligent decision support unit to perform any preventive actions, monitor manufacturing process and inspection thus enhancing resource efficiency, product quality and production efficiency by 95%.
- Enhanced defect recognition software, and image processing techniques/algorithms for both optical inspections and NDT techniques enabling to analyse and detect 100% defects in variety of optical devices.
- Innovative recycling/re-use and recycling approach allowing production loss reduction by 40% due to re-use and repair of rejected components and devices.

Scale Up: The project objectives have been defined considering scaled up approach for mass production lines, to manufacture, inspect and monitor at every stage of process, more than one system can be deployed to monitor different stages of development of opto-electronic devices.

OLEDSOLAR will give Europe a technological and competitive advantage in the growing opto-electrical manufacturing and production industry. The key factors of project scale, challenge, and depth of required technical skills demand the work to be pursued as **multi-actor industrial approach through a collaborative project** bringing together European expertise in aspects related to manufacturing processes, inspection systems, metrology, process automation, quality control, NDT, opto-electrical components production companies and recycling expertise as mentioned below:

16 partners, 7 RTOs, 1 University, and 8 industrial partners including large companies and SMEs,

Industrial/Companies Partners:

- **ASPF** (End-user OPVs: Business/Industrial needs requirements + Organic photovoltaics production line)
- **Solibro** (End-user CIGS: Business/Industrial needs requirements + CIGS thin-film modules production line)
- **INURU** (End-user OLEDs: Business/Industrial needs requirements + (OLEDs production line)
- **Advent** (Recycling: Ink formulation for OPV and OLED manufacturing + Testing novel materials for recycling)
- **DuPont Teijin Films** (Process: Manufacturer of films for deposition substrate)
- **Smit Thermal Solutions** (Equipment supplier: spatial and plasma enhanced ALD equipment developer and test bench manufacturer for R2R pilot line)
- **IRIS** (In-line metrology: Implementation and deployment of optical characterization techniques (e.g. diffractometry, automation and sensors system for process control)
- **Coatema** (Equipment manufacturer, roll to roll and S2S processes scale up, system integrator etc)

Research Organisations:

- **VTT** (Manufacturing processes optimisation, testing bed: R2R testing pilot line (R2R processing of OLEDs, OPVs, inline monitoring during printing, printing quality measurements, printing machine monitoring) + Testing novel processes for recycling)
- **LEITAT** (Recycling: Testing novel materials for recycling)
- **TWI** (In-line metrology: Implementation and deployment of NDT techniques for device characterization (e.g. laser shearography, illuminated lock-in thermography))
- **Fraunhofer** (barrier film qualification, testing OLED devices pilot lines, inline metrology)
- **TNO** (Spatial and plasma enhanced ALD process optimisation for OLEDs and Solar cell manufacturing)
- **CSEM** (Manufacturing processes, metrology, S2S fabrication environment, measurement techniques etc.)
- **CTR** (In-line metrology: Know-how in Raman spectroscopy for material characterisation + luminescence measurements of PV cells and encapsulants)

University Partners:

- **BUL** (AI-process control: Data analysis provider (AI-driven techniques for device characterization and roll-to-roll process control))

1.2 Relation to the work programme

The project relation with topic expectation is detailed in table below:

Table 1: Project relations with topic scope

Work programme Call - FACTORIES OF THE FUTURE - FOF

Expectation from Work programme	OLEDSOLAR Project addressing the expectations
The Factories of the Future Public-Private Partnership (PPP) initiative aims at helping EU manufacturing enterprises, SMEs, to adapt to global competitive pressures by developing the necessary key enabling technologies to support EU manufacturing across a broad range of sectors. It will help European industry to meet the increasing global consumer demand for greener, more customised and higher quality products through the necessary transition to a flexible, digitalised and demand-driven industry with lower waste generation and energy consumption.	OLEDSOLAR project is in line with FOF PPP initiative as it is aimed at improving competitiveness of manufacturing industry involved in development of opt-electronic devices e.g. OLEDs, OPVs and CIGS by offering and developing enhanced manufacturing process, improved metrology and automated inline measurement, control and inspection systems capable of enhancing production yield and analysing 100% of defects during production of components at production rate. OLEDSOLAR will help EU industrial sector to meet consumer demand in terms of high quality manufacturing, 100% reliable and efficient products and components with minimum waste generation and efficient resource utilisation.
Call Topic: DT-FOF-03-2018: Innovative manufacturing of opto-electrical parts	
Topic Challenge	**How the project addresses the Challenges**
New adjustable processes for scaled up production of free form and complex opto-electronic parts and devices and equipment for testing, failure analysis and control using range of sensors.	OLEDSOLAR proposed manufacturing processes, inspection techniques and tools, automation through sensors network and recycling processes development will address the challenge of opto-electronic parts and devices manufacturing, process adjustment as per devices customisation and specification needs and defect free manufacturing of such parts in production through inline monitoring.
Scope of the topic	**How the project addresses the scope?**
New, flexible, and innovative process chains to handle complex designs that include opto-electrical functionalities,	OLEDSOLAR project proposes innovative manufacturing processes including automated printing techniques for OPVs allowing free form module production and devices customisation. Further, innovative deposition techniques proposed will improve CIGS and OLEDs devices performance and efficiency by enabling enhanced barrier film performance, the proposed inline inspection and testing techniques (photoluminescence, optical diffraction, Raman spectroscopy, spectroscopic ellipsometry) that will result in defect free production of OLEDs, OPVs and CIGS devices.
Improved sensor equipment for quality control in the different processing steps as well as the final functionality of the component,	OLEDSOLAR will implement improved and optimised network of sensors and testing equipment for quality control and monitoring of different manufacturing processes of OLEDs and solar cells.
Methodologies for improving quality through high-precision automation using the sensor data, including non-destructive in-process evaluation of material and functional component properties,	OLEDSOLAR project proposed methodologies will allow precise automation enabled by sensors and intelligent decision management software for production of OLEDS and solar cells within R2R and S2S production environment. For NDT and in-process evaluation of material and functional properties of OLEDs and solar cells during production; development of hybrid inspection and measurement techniques and systems will be optimised in OLEDSOLAR including Photoluminescence, optical diffraction, shearography, thermography, Raman spectroscopy, Spectroscopic ellipsometry, HST and OCT etc.
Re-use and requalification of key components and precious materials within the process chain from products at their end of life,	Recycling and re-use approaches will be developed in the project allowing more than 40% of defected part to be reused and recycled.

1.3 Concept and methodology

(a) Concept : The overall project objective is twofold: (1) to design, build and scale roll-to-roll manufacturing process and technologies (e.g. atomic layer deposition) aimed at complete protection of solar cells and OLEDs against humidity, higher degree of automation, higher throughput and reduced cost, etc. Project results will be integrated and demonstrated in a pilot line for continuous and high-volume production of CIGS photovoltaics; (2) to design and implement, in two pilot lines, a real-time quality monitoring process based on hybrid optical measurement techniques, aimed at monitoring, controlling and improving the quality of the production of optoelectronic organic devices. The two pilot lines will reflect two wide industrial areas facing up-scaling challenges: OLED industrial market (organic lighting panels), building integrated organic solar cells and CIGs.

The project is comprised on four main pillar technologies:

1. Development of innovative manufacturing processes for OLEDs and solar cells (in R2R and S2S)
2. In-line quality inspection and testing of OLEDs and solar cells
3. Recycling and Re-use approaches
4. Process Automation and control

The baseline for overall concept and methodology of proposed OLEDSOLAR solution is comprised of following:

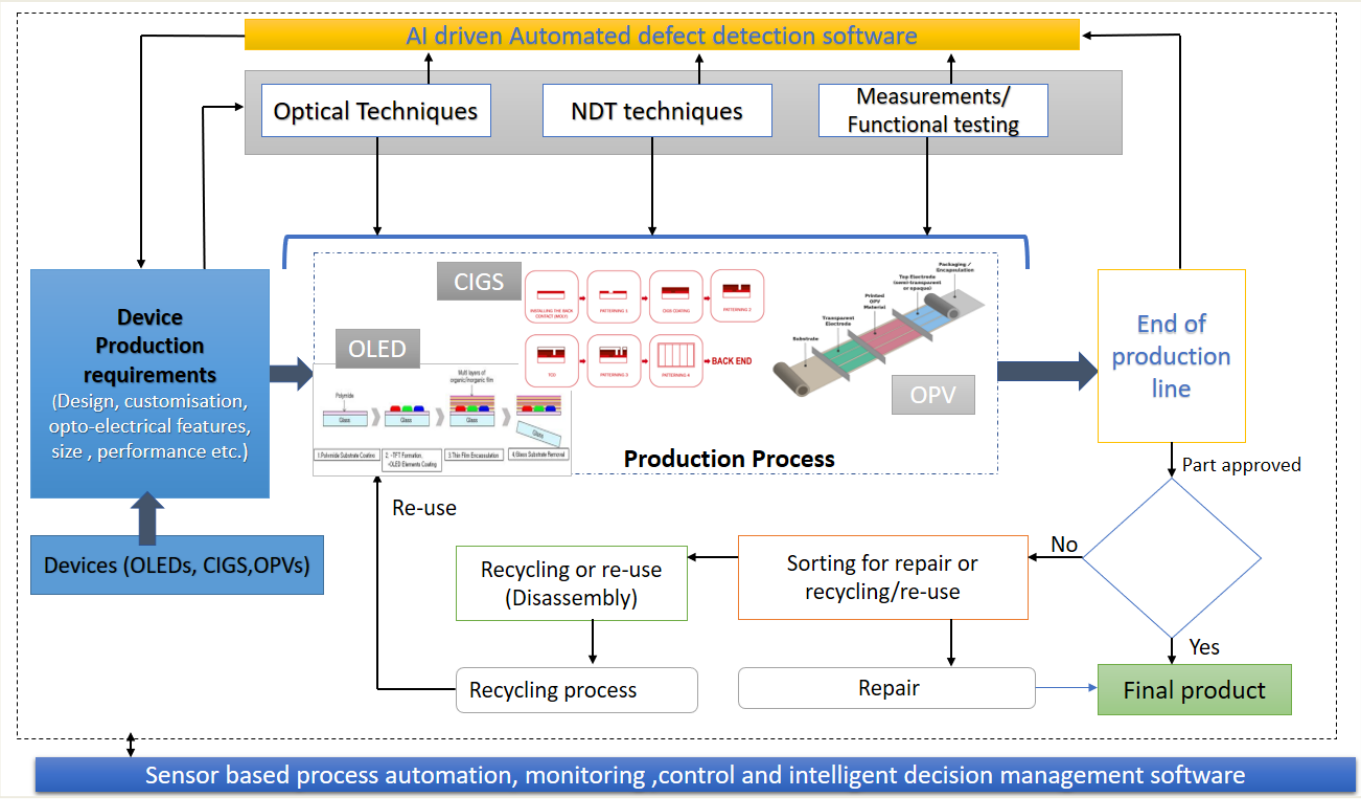


Figure 4 Project concept layout

OLEDSOLAR project proposes OLED and (CIGS and organic) solar cells manufacturing processes within R2R and S2S manufacturing environment including pattering, deposition, packaging, lamination handling techniques and systems. Most of the processes being developed are at TRL4 with few at TRL5 and project objective is to reach TRL6. The validation of these processes will be done for specific device in pilot lines and production lines. **(Activities planned in WP1,2,6)** Further, a hybrid system of number of inspection and testing techniques and equipment **(Activities planned in WP3,6)** will be optimised in OLEDSOLAR to detect, control, and otherwise eliminate potential quality issues within the production lines including;

- Mechanical defects (e.g. Shearography): Inspection for mechanical defects such as pinholes and cracks;
- Electrical-optical properties (e.g. Photoluminescence, Photo-CELIV, Modulated Electroluminescence Spectroscopy, Impedance Spectroscopy, Transient Electroluminescence): Monitoring of opto-electrical properties such as current, voltage, luminance, mobilities and the phosphorescence lifetime in an OLED, etc;
- Interfacial behaviour (e.g. Spectroscopic ellipsometry): Thermo-mechanical inspection for evaluation of interfacial integrity in multi-layered structures (interfacial delamination); monitor interface thickness in real-time: film growth, adsorption; monitor real time changes at interfaces;
- Surface measurement (e.g. Spectroscopic ellipsometry): Roughness thickness, native oxide thickness, depolarization coefficient;
- Thickness measurement (e.g. Spectroscopic ellipsometry): Single and multi-layer thickness measurement;

- Material properties (e.g. Spectroscopic ellipsometry): Film porosity, gradient, microstructure information
- All inspection and data acquisition systems would be integrated into process control and feedback systems throughout the production line. These technologies will be used to correlate defects to performance (lifetime and efficiency);
- Barrier film qualification and HSI
- OCT technique for benchmarking and validation,
- Measurement tools to monitor manufacturing parameters e.g. printing speed etc.

Moreover, recycling and re-use strategies and processes will be developed to dismantle discarded devices and sort out parts to be re-used or recycled into materials and fed back into production processes. Repair strategies will also ve developed in OLEDSOLAR project**. (Activities planned in WP5,6)**

The whole production process will be automatized using network of sensors and intelligent decision management software, **(Activities planned in WP4,6)**

The overall project scope is shown in figure below:

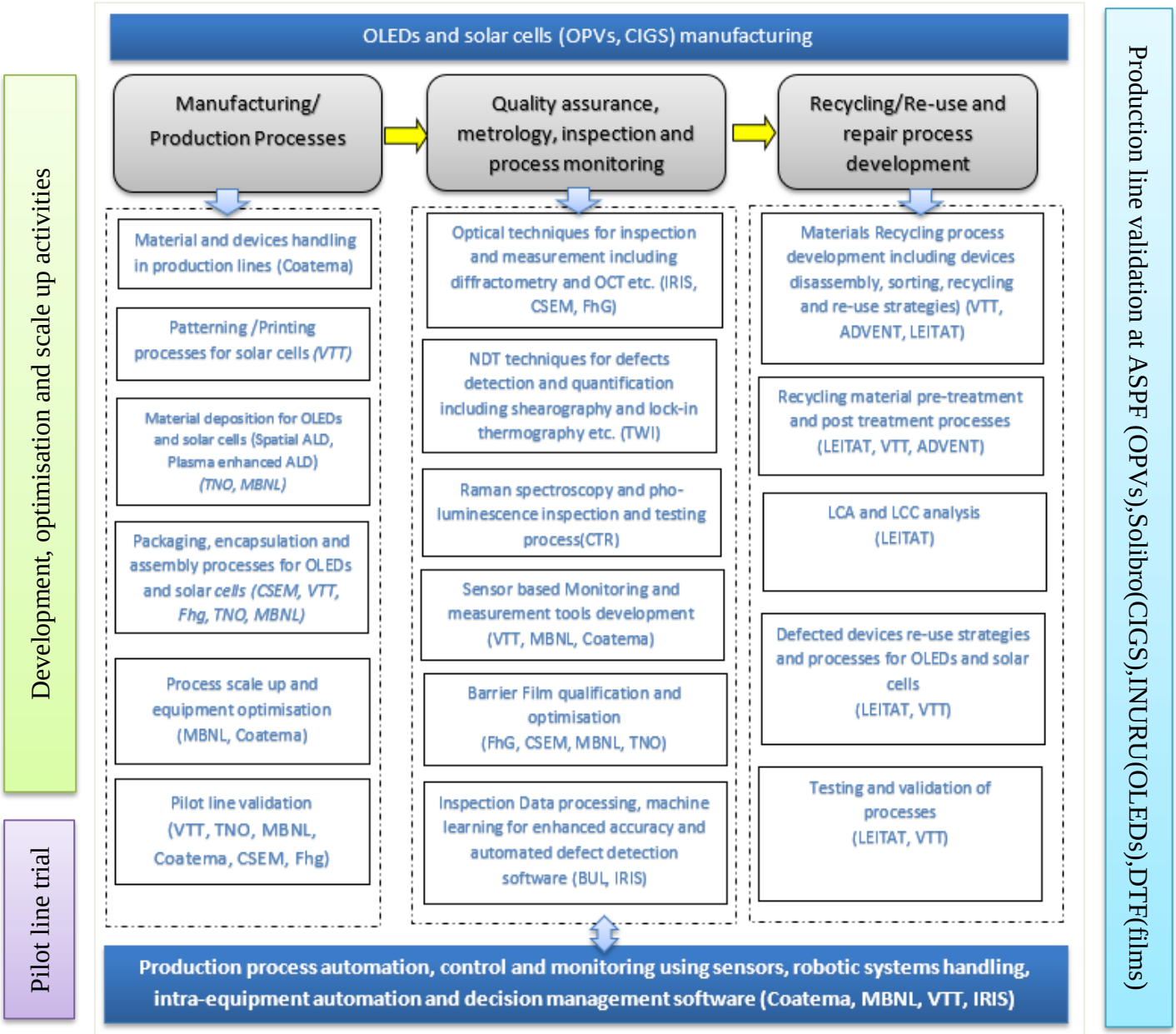


Figure 5 Project activities detailed scope and partners involvement

Overall chain of project development:

1. Development of innovative manufacturing processes for OLEDs and solar cells

Several recent developments in material science and deposition methods for flexible organic light-emitting devices (OLEDs) are surveyed. The commonly used plastic substrates are compared, according to their mechanical, optical, thermal, and chemical properties. Multilayer electrode structures, used as transparent electrodes, replacing conventional indium tin oxide (ITO) are presented and data about their conductivity, transparency, and bending

ability are provided. Attention is paid to some of the most popular industrial processes for flexible OLEDs manufacturing, such as roll-to-roll printing, inkjet printing, and screen printing. Main specifics and challenges, related to the foils reliability, mechanical stability of the transparent electrodes, and deposition and patterning of organic emissive films.

Material handling: Material handling embraces all of the basic operations involved in the movement of bulk, packaged, and individual products in a semisolid or solid state by means of machinery, and within the limits of a place of factory and production line. The objectives of material handling are further defined as:

- Increase efficiency of material flow
- Improve facility utilization
- Improve safety and working conditions
- Facilitate the manufacturing process
- And all the above at a reduced cost

Material handling can account for 30-75 percent of production costs and can reduce operational costs by 15-30 percent (Frost and Sullivan). In OLEDSOLAR project, pick and play, lamination tool and substrate handling system will be optimised up to TRL6 on the basis of development already achieved by Coatema (at TRL4).
Coatema has developed and demonstrated the Click&Coat® Technology for R2R-pilot and production lines (Figure 8). This means that individual modules, e.g. winder, edge guiding, corona, coating, dryer or lamination for each process have been developed. Bespoken modules, e.g. inert coating, rinsing, registration and cutting have also been developed. With these modules any process can be configured, e.g. with minimum front-site contact. You only use the modules needed for the desired process. Within 1 h the configuration of the Click&Coat®-modules can be changed to enable a totally different production process. As a consequence, several production steps (e.g. to address niche markets) can be realized with only one pilot line. Therefore, the Click&Coat®-Technology offers the highest equipment flexibility with a high utilization and low cost of ownership.

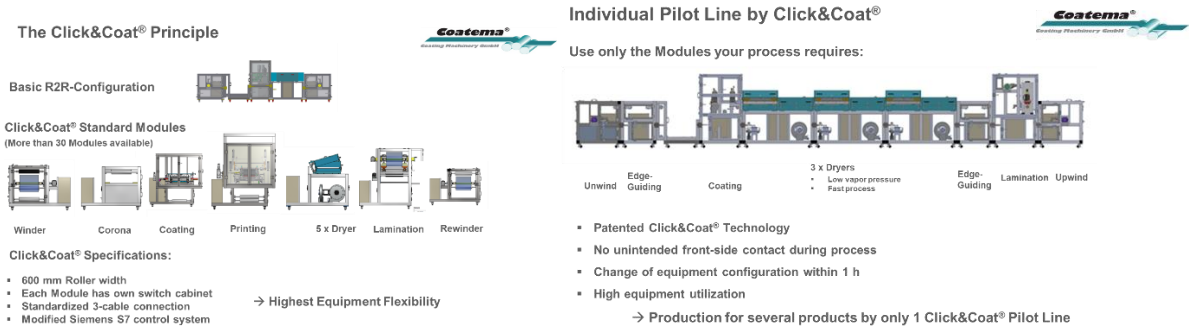


Figure 6 The principle of Click&Coat®-Technology by Coatema

Figure 9 shows a coating line, which has been developed in the EELICON-Project. This Click&Coat®-configuration consisted of 12 individual modules resulting in a total length of 23m.

Figure 7 The Click & Coat®-Pilot Line at Coatema.

Figure 8 Coatema´ s Engineering solution for inert slot-die coating (left), lamination, LED-curing (center) and its final cutting (right), which was realized in the EELICON-project for the electrochromic film R2R-Pilot line.

Such adaptable click and coat system will be optimised in the OLEDSOLAR project considering validated above system for electrochromic films are similar in its process requirements as OLEDs or OPV. The film thickness and thickness homogeneity are in the range of nm. The films are sensitive towards moisture and particle contamination. Within the OLEDSOLAR consortium it can be expected that similar cost saving predictions by novel engineering

solutions can be anticipated in the engineering study from Coatema for the existing pilot lines. *This system will contribute to the impact of process reconfiguration time by achieving* ***time reduction by more than 15%.***

Manufacturing processes at CSEM

1) **Semi-transparent OPV module** A typical process flow consists of the following steps: 1) Design of the module. Using an advanced FEM solver both the number and the size of interconnected cells are optimized for maximum power output. To reduce resistive losses the ITO bottom electrode is metallized with a metal grid with geometry optimized for low resistivity. 2) Generation of the layouts of the layer in the module, including alignment marks, 3) ITO patterning by chemical etching or laser ablation, 4) Thermal evaporation of the metal on the ITO, 5) blade or slot-die coating of the layer, 6) Patterning of the coatings into individual cells, 7) Deposition of the metal top electrode (by thermal evaporation, or screen printing).

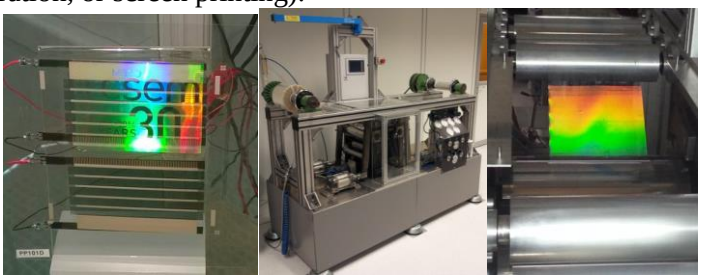


Figure 9 (a) Printed flexible OPV module. The top module includes a nanostructured foil with the CSEM logo. The foil is embedded between the OPV module and the outer barrier. The nanostructures were designed such that the logo becomes visible under a steep angle

2) Diffractive nanostructure

1) Design of the structures using rigorous coupled-wave analysis (RCWA), 2) Origination of the structures using laser interference lithography, 3) Fabrication of first or higher generations of Ni shims by galvanic growth. 4) Creation of a stamp by UV replication, 5) Upscaling to 20cm x 40cm using automated step & repeat UV replication, 6) Fabrication of large area Ni shim, 7) Large area shim is mounted on hot embossing pilot line, 8) Hot embossing into thermoplastic polymer films.

Material deposition:

Spatial atomic layer deposition:

Atomic layer deposition (ALD) technique is a well acknowledged thin film growth technique to generate high-quality materials such as oxides, nitrides, sulphides, and metals. ALD method is based on the sequential exposure of a substrate to self-limiting surface reactions, which enable a sub-nanometre scale thickness control of the deposited material. By the self-limited nature of these reaction, ALD allows the fabrication of continuous, dense and virtually pinhole-free layers with an excellent surface coverage and unrivalled conformality also on complex structures. Whilst ALD is already commercially adopted in the semiconductor industry (i.e. high-dielectric oxide for field effect transistors), the benefit of this technique has been successfully explored also in other applications such as flexible displays, flexible organic light- emitting diodes and photovoltaics. ALD Al_2O_3 layer works as an excellent surface passivation layer in crystalline silicon solar cells. Zn(O,S) based buffer layers in copper indium gallium selenide (CIGS) are alternative candidate to cadmium sulphide films towards cadmium free CIGS solar cells. Finally, several ALD layers have been proven as excellent moisture diffusion barrier films for OLEDs and thin film photovoltaics. Despite the superior quality of these functional ALD layers, the conventional ALD technique is not economically compatible with all the applications where high-throughput and low-cost in line production techniques are required. Indeed, conventional ALD processes exhibit very low deposition rate in the order of A-nm/min, requiring expensive vacuum systems which add complexity when upscaling to large substrate sizes. In order to meet the requirements of fast depositions and low-cost manufacturing processes, spatial ALD (s-ALD) approach has been developed. In the conventional ALD, the precursors are dosed inside the reaction chamber in two different moments, separated in time by purging steps (Figure 1). Spatial ALD approach relies on the injection of the precursors in two separate zones of the reactor, using inert gas curtain to provide the physical barrier separation and to avoid any precursor mixing and thus parasitic chemical vapour deposition contribution in the film growth. Since purge times are no longer required, s-ALD enables to significantly increase the deposition rate, which become only dependent on how fast the sample can be moved, on the design of the reactor, and ultimately, on the kinetic of the surface reactions. Due to the presence of the gas curtain, the reactions zones are completely isolated from the external environment, enabling operation under atmospheric pressure conditions. The last, is extremely appealing since make the spatial ALD technique compatible with other in line low cost deposition technique such as ink jet and slot die printing.

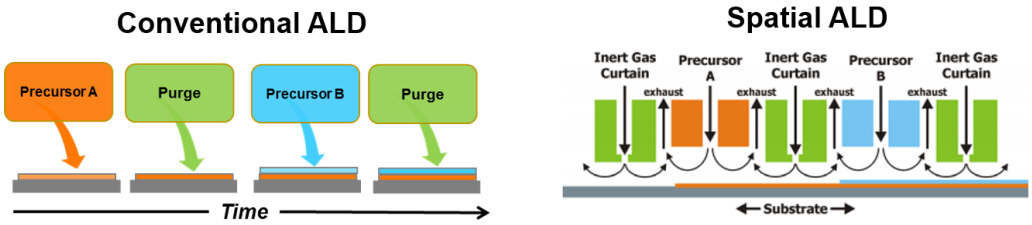


Figure 10 Schematics representation of a) conventional ALD process where the separation of the precursors in time is provided by the use of purge steps, b) spatial ALD process where the separation is ensured by the injection of precursors in two different zones of the reactor and by the presence of inert gas curtains

TNO combined gas bearing technology with atmospheric pressure gas phase deposition technology. The use of gas bearing enables to minimize the distance between the reactor and substrate with an accurate control of several tens of micrometres. The first concept of spatial (S-)ALD lab scale tool was developed in 2009 (See Figure 2). A second one has been installed in 2016. Both tools can be run with the injector head with a plasma source, enabling atmospheric plasma enhanced (PE) (S-) ALD.

TNO developed also a substrate contactless roll-to-roll spatial ALD reactor consisting of a central drum that contains one or more combinations of metal precursor and oxygen source (i.e. water vapour) half-reaction zones. These zones are separated and surrounded by nitrogen gas bearings. The foil to be coated is pulled over this drum, where the gas bearing ensures that the foil is kept at a fixed distance from the surface of the drum (typically tens to hundreds of micrometres). The absence of mechanical contact between the deposition side of the foil and the reactor minimizes contamination of the foil with particles which would lead to detrimental pinholes in the deposited films. High deposition rate can be achieved moving the foil in one direction and rotating the spatial ALD injector drum to the other one. A prototype has been installed at TNO with a web width of 30 cm.

TNO developed also a sheet to sheet (S2S) Spatial ALD pilot line to process substrates size of maximum 325 x 400 mm. This tool can also be run with an injector tool with plasma source.

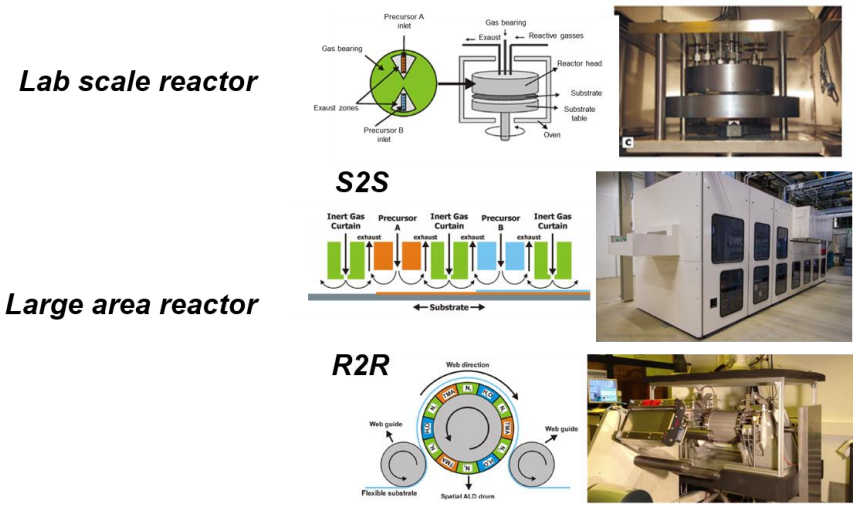


Figure 11 Schematic representation and image of lab scale, large area sheet to sheet (S2S) and roll to roll (R2R) s-ALD reactors.

TNO is currently developing new tools, processes, based also on atmospheric plasma technology, with the aim of upscale these solutions on S2S and R2R tools for applications such as encapsulation, photovoltaics, and solid-state lighting. SMIT TS has engineered both R2R and S2S tools that perform Spatial ALD. For pilot production on flexible substrates, the FLEx R2R sALD system (Figure 11) is available for coating on foil up to a web width of 60 cm. Currently the ALD process uses TMA and water for deposition of AlOx functional barriers, whereby the H2O is thermally decomposed. These tools are presently at TRL 5(“ . The foreseeable challenges on the path to TRL6 are:

- establishment of standard operating procedures and maintenance (WP2 & WP4)
- meaningful characterization of several hundreds of square meters (WP3)
- stability of web height over several hours and its gas separation performance(WP2 & WP4)

Figure 12 FLEx Roll to roll (R2R) s-ALD pilot production system for a web width of 60 cm made by SMIT TS.

The Plasma enhanced spatial atomic layer deposition has been developed and implemented on the S2S sALD tools. For the R2R systems, this is new and is desired for additional

flexibility of precursors (e.g. water–free oxidant) and process window (e.g. low temperature deposition) and compatibility with moisture sensitive devices. is an option on the R2R and S2S sALD tools. Its maturity is TRL4. The challenges for this specific technology are:

- Development, integration and testing of PE sALD compatible with R2R pilot production
- Understanding of the plasma interaction with the substrate (WP3)
- Relation between quality of the layer and (plasma) process parameters (WP2)

Process goal is a low temperature deposition on moisture sensitive devices on flexible web.

The technology, now at its TRL4 (" , Plasma Enhanced Spatial Atomic Layer Deposition (PEsALD), is applied in a Roll to Roll (R2R) equipment.[10]

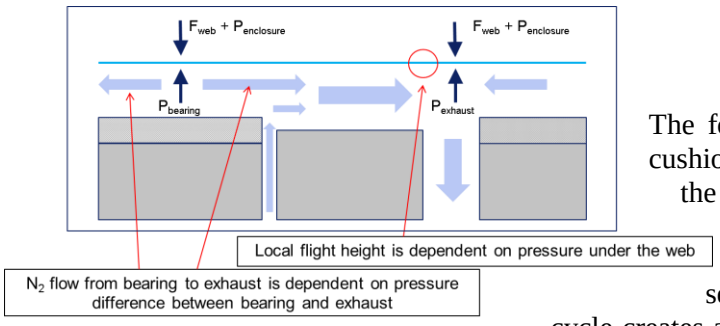


Figure 13 PESALD Process layout

The foil is brought on a drum where, by means of a gas cushion, floats without touching it. In the clearance between the drum and the foil the deposition takes place. The deposition occurs when the substrate is exposed to alternating precursor gases, usually two. This separation is achieved by an additional bearing gas. Each cycle creates a layer in the crystal structure. Hundreds of cycles are needed to achieve few tens of nanometre.

For the case of an Aluminium Oxide deposition these are the three gases of choice:

- material precursor gas 1: Trimethylaluminium (TMA),
- Material precursor gas 2: Water damp (H_2O), O2, O3
- reaction confinement purging gas: Nitrogen (N_2)

Due to the sensitivity to high temperature and moisture of the targeted surfaces, alternative low temperature processes are desired. This can be achieved by using a plasma to dissociate the oxidant and makes possible using alternative precursors for oxidation that avoids direct exposure to moisture, of important for moisture sensitive devices. (Langereis et al. 2006). A test bench (or loop-coater) will be developed to facilitate integration of a plasma enhanced (PE) unit compatible with an R2R sALD system. This needs to be developed as it is non-existent at this moment and adds a high degree of innovation to the project. Aside for allowing integration and testing, the test bench will make possible coating of foil with and without PE for analysis of layer functionality and product compatibility (half-fabricates from partners). The developed unit is meant to be compatible with an existing R2R sALD pilot production system to allow for end-users to select such a system for a (future) pilot line.

Material deposition: A thin (5nm to 50nm, preferably 20nm) Aluminium Oxide layer is deposited conformally on the devices. This layer has excellent moisture barrier properties ($<10^{-4}$ g(water)/m^2/day). Such deposition can be made more robust by using organic pre-coating and post-coating respectively for planarization and mechanical protection purposes.

Production system: The functionality that the AlOx layer delivers is qualified by its long-lasting quality. Testing this function therefore means verify for a long time that the layer is performing. This is not in line with a quick feedback. For this reason, proxy measurements are used.

- geometrical and composition changes of the layer (spectroscopic ellipsometry)
- particle or irregularities on the substrate (electrical optical properties)
- cracks initiation (shearography)
- others

These proxies are established by experience; however, their effectiveness is at TRL4 stage.

Process adjustments have to be both rapid and accurate:

Organic Pre and post coating could be digitally defined by using an inkjet printing platform. Furthermore, the plasma enhancement can open structuring freedom[11].

2. Inline In-line quality inspection and testing of OLEDs and solar cells

Monitoring system for the organic thin film deposition process in OLEDSOLAR

Several optical monitoring systems exist that have a suitability to different coating features (dry/wet, transparent or not, different thickness, etc). The project approach will be to combine and tailor commercially available technologies

[10] (Kessels and Putkonen 2011; Poodt, Cameron, et al. 2011).

[11] (Poodt, Kniknie, et al. 2011)

for the target application and some have already been implemented successfully by IRIS in previous projects for OPV lines. Specifically, two optical nanometrology monitoring systems, one based on spectral reflectance and another on ellipsometry will be designed and built for measuring continuously and in real-time thin film thickness and/or yield. These sensors will be capable of measuring thin films on polymeric substrates either below a micron or in the tens of microns range offering a rapid response. Dispersion will be tested offline during process optimization and is not expected to vary once robust processes are established although their impact on the final properties such as barrier would be detected by the quality control tests of the products. Nevertheless, some online thickness measurement down to the nanoscale will be integrated in the project as one of the most important acceptance criteria for the industry for the nano-enhanced processes to warranty constant barrier and effect of the surface modification for the products resulting from the upgraded processes. The sensors will be integrated in a suitable test rig to enable their off-line calibration vs. standard measurements using relevant sample sets selected by the end users. The previously built monitoring systems will be integrated in the respective target production lines for in-line thickness measurement.

Advances in thickness measurements of organic coatings and deposition processes in OLEDSOLAR

Optical techniques are the preferred method for measuring non-opaque thin films because they are accurate, non-destructive, and require little or no sample preparation. The two most common optical measurement techniques are spectral reflectance and ellipsometry. Spectral reflectance measures the amount of light reflected from a thin film over a range of wavelengths, with the incident light perpendicular to the sample surface. Ellipsometry is similar, except that it measures reflectance at non-normal incidence and at different polarizations. In general, spectral reflectance is much simpler and less expensive than ellipsometry, but it is restricted to measuring less complex structures. In view of the importance of nanometrology for obtaining fast product and process design, different optical systems based on spectral reflectance and spectroscopic imaging ellipsometry will be used for high speed inline monitoring of the coating application processes (range 1-50 microns dry) and, on the other hand, to assess the quality of the deposition process (range expected 50-200 nm dry) on a range of substrates.

System based on spectral reflectance: Reflection occurs when light crosses the interface between different materials. The reflections from the different interfaces may add together either constructively or destructively depending on their phase relationship, which is determined by the difference in optical path lengths of the reflected beams, which in turn is determined by thickness of the film, its optical constants, and the wavelength of the light. Therefore, the reflectance of a thin film will vary periodically with the inverse of the wavelength, and thicker films will exhibit a greater number of oscillations over a given wavelength range, while thinner films will exhibit fewer oscillations or only part of an oscillation, over the same range. Spectral reflectance can be used to measure the thickness, roughness, and optical constants of a broad range of thin films. However, if there is less than one reflectance oscillation (i.e. the film is very thin), there is less information available to determine the adjustable model parameters. While the number of film properties that may be determined decreases for very thin films, if only thickness is of interest the minimum single-film thickness that can be measured using spectral reflectance is about 10 Å to 300 Å. However, when solving for the optical properties of more than one film, the minimum thickness increases further[12].

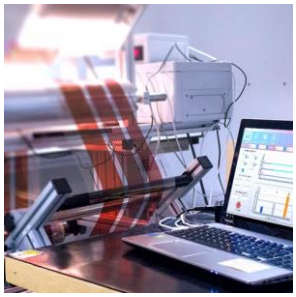


Figure 14 Opti nano Pro system

Spectral reflectance can be used to measure a large number of technologically important films. For example, it is widely used in the semiconductor industry to measure the thickness, roughness, and optical constants of oxides, SiNx, photoresist, and other semiconductor process films. It is also used to monitor thin films used for scratch resistant and antireflection coatings, automotive plastics, eyeglass lenses, and many plastics packaging applications. However, when films are too thin, too numerous, or too complicated to be measured with spectral reflectance, they can often be measured with the generally more powerful technique of spectroscopic ellipsometry.

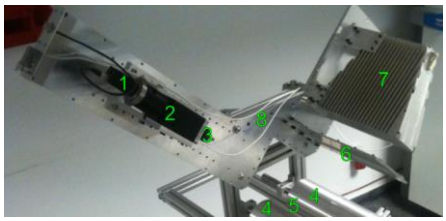


Figure 15 Hyperspectral imager

System based on ellipsometry: Ellipsometry measures a change in polarization as light is reflected from a surface. When linearly polarized light is reflected from a surface, in general the amplitudes of its two perpendicular electric field components change and so does the phase between them. The light becomes elliptically polarized. Various types of ellipsometry exist, but a common form is the rotating polarizer. In this design, linearly polarized light is incident on the sample surface, but the polarizer is rotated, so that the plane of polarization rotates about the axis of the incident light. On the detector side, the (generally) elliptically polarized light is passed through a fixed analyser. The intensity of light on the detector oscillates with rotation of the incident polarizer according to the amplitude and phase change. The polarization change

[12] Filmetrics – Thin-film measurement technical note

is represented as an amplitude ratio and a phase difference. The measured response depends on optical properties and thickness of individual materials and therefore ellipsometry is primarily used to determine film thickness and optical constants[13]. By measuring reflectance at non-normal incidence, ellipsometry is more sensitive to very thin layers, and the different polarization measurements provide more information for analysis. The trade-off here is a more complex and costly system as well as a slower speed of operation. Online monitoring module based on ellipsometry to control the deposited thickness of very thin graphene films was already reported[14].

Figure 16 Ellipsometry system

Online spectroscopic ellipsometry has been implemented on roll to roll printed organic electronic layers[15], and commercial prototypes have been developed by Horiba[16] and Semilab[17].

There has been extensive work on metrology of nanoparticles using ellipsometry, making use of effective medium approximations to model the structured layer as a combination of its constituent parts. Thickness, optical and electronic properties have been measured for a variety of nanoparticle layers including SiO2 particles in ORMOCER barrier layers on PET[18], HgTe optoelectronic nanocrystal multilayers[19] and gold plasmonic nanostructures[20]. Part of the advancement of the state of the art in OLEDSOLAR will be to apply thickness measurement using the above-mentioned technologies inline on organic thin film deposition processes.

Figure 17 Inspection data measurement software

Raman & Luminescence In-Line Metrology

The fundamental feasibility of investigating of organic LEDs and organic PV components and systems by means of Raman (micro-)spectroscopy (RS) and/or different luminescence techniques, in particular photo-luminescence (PL) and electro-luminescence (EL), is well established. It has been shown that RS can be used to identify and quantify different components, like fullerenes in polymer matrices that constitute a vital of organic photovoltaics.

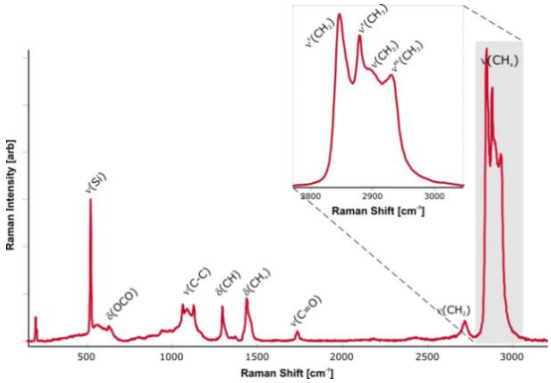


Figure 18 Typical Raman spectrum of a cross linked ethylen vinyl acetate (EVA) sample.

Equally, RS can distinguish between different crystal morphologies, both for organic and inorganic materials as well as measuring mechanical stresses and stress distributions. Exploiting phonon effects, it is even possible to non-destructively measure dopant concentrations and their electrical effects in materials, a capability that is of interest e.g. for studying CIGS layers and the like. The limitation with regard to the objectives of OLEDSOLAR is that almost all of this studies focus on the material science part, aiding the development of new (nano-)materials and studying light-material interaction. While RS is now being routinely used as a PAT tool in e.g. the pharma industry, very few examples of true RS process analysers have been described for LED and/or PV manufacturing. It has, however, been shown that RS can be used as an in-line tool, for instance to measure the degree of

Figure 19 Characterization of defective photovoltaic modules by using electroluminescence imaging (Source: CTR)

[13] The complex refractive index and reflectometry versus ellipsometry. N J Elton. Surfoptic Reflectometry Technical Paper No. July 2007
[14] http://spie.org/documents/Newsroom/Imported/005227/005227_10.pdf
[15] S. Logothetidis et al. / Solar Energy Materials & Solar Cells 112 (2013) 144–15
[16] http://www.horiba.com/fileadmin/uploads/Scientific/Documents/TFilm/uvisel_inline.pdf
[17] P. Basa, Lope-C 2013, In-line thickness and composition monitoring of deposited coatings in a roll-to-roll process tool by spectroscopic ellipsometry
[18] A. Laskarakis et al. / Thin Solid Films 517 (2009) 6275–6279
[19] V. Rinnerbauer et al. / Physica E 32 (2006) 104–107
[20] M. Losurdo / Thin Solid Films 519 (2011) 2575–2583

crosslinking in assembled (classical, i.e. Si cell-based) PV modules. The situation is similar for PL and EL, which, however, have recently been shown to be applicable also for investigating modules in the field.
EL measurements take advantage of the radiative inter-band recombination of excited charge carriers in solar cells. For EL investigations the module is operated as a light emitting diode and the emitted radiation due to recombination effects can be detected with a sensitive CCD camera. The solar cells are supplied with a defined external excitation current (current applied=ISC) while the camera takes an image of the emitted photons. Damaged areas of a solar module appear dark or shine less than areas without defects.
Electroluminescence imaging can be used to detect a number of defects in crystalline and thin film solar cells. The high resolution of the EL-images enables to resolve some defects more precisely than in IR-images. It was shown that electroluminescence is a good method to detect micro cracks, interrupted contacts or a number of process failures (e.g. shunts or defects in the antireflection layer) but it was not possible to determine the influence of these defects on the cell/module power output.
The first challenge and expected advance of this task of OLEDSOLAR over the state of the art is hence to match the expectations by the component and systems manufacturers with inherent spectral and/or spatial features, to establish the basic feasibility of the required analysis by one or more of the aforementioned methods. The second step is then to select features that are sufficiently stable to have a chance to yield reliable results under process conditions; this involves, for instance, investigating approaches for self-referencing models, both along the spectral scale (to eliminate e.g. thermally induced spectral drifts and minimise otherwise necessary efforts for extremely reliable temperature conditioning of the spectrometers) and the intensity information. The objective behind this is to create robust models that have a realistic potential of conveying meaningful information on the process or product in real-time under process conditions. The third R&D phase then deals with a test implementation on an industrial scale, involving customised sample-device interfaces, relying e.g. on fibre optics, dedicated algorithms and spectral analysers customised to the spectral features of interest to maximise both reliability and analytical sensitivity.

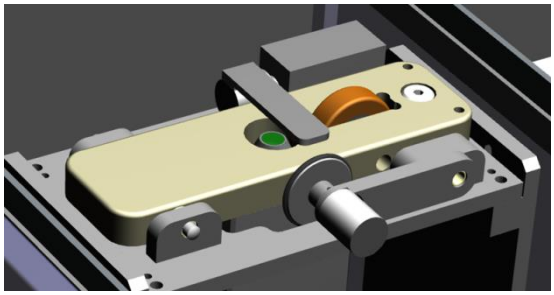


Figure 20 schematic sketch of a inline PV quality measurement setup with Raman inspection head (Source: CTR)

Hyperspectral imaging

Hyperspectral imaging (HSI) combines spectroscopic analysis with and imaging technologies. Dedicated cameras enable the simultaneously acquisition of spectral data at many sample positions from the UV up to NIR. Different data analysis approaches enable the spatial resolved inspection of certain sample properties (e.g. optical properties, properties of the sample topology) or the evaluation of properties of a certain sample area (e.g. its quality). On one hand the theory of beam propagation in layered systems will be used on the other hand multivariate data analysis and deep learning methods are used. The latter are in particular the basis for inline application of HSI technology in combination of data acquisition in the range of kHz

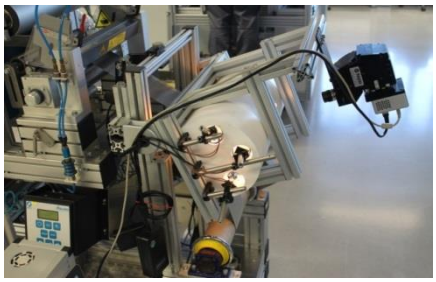


Figure 21 Hyperspectral imager at a R2R-process

Inline NDT techniques:

Shearography has been used in industry for defect detection to supplement ultrasonic and other NDT techniques. Digital shearography is one of the most promising techniques, with the potential for rapid non-destructive evaluation as well as the quantitative measurement of strains, stresses and vibrations. Compared with traditional NDT techniques, shearography reveals a defect within an object (e.g. organic solar cell device) by identifying defect-induced deformation anomalies. The commonly used means of stressing are heating, vacuum-pumping (by a vacuum chamber or a vacuum hood) and vibration excitation. Digital shearography is effective in detecting a variety of flaws such as voids, disbonds, delamination etc. However, flaw characterisation in practice is a complex task, which depends on both the nature of the flaw (shape, size, location) and the status of the applied stress at the location of the flaw. The key to the success of shearography in defect detection is to impose an appropriate stress status on the object under investigation. If an appropriate means of stressing is adopted, shearography can inspect a relatively large area in a fraction of a second.

Experimental procedure:

- The laser shearography principle consists of interferometric comparison of a test object in two states, an initial and a loaded one, or at different loading conditions;
- Use of thermal heating as the means of stressing the samples for defect detection (e.g. by using a heat gun to heat the samples for a short period of time, during which shearography fringe patterns are acquired in real time or through post processing of the recorded laser speckle images);
- For thin panels or film components, vibration excitation may be a better stressing means. A PZT actuator may be used to produce vibration excitation. With time averaged shearography, it is expected to detect delamination and/or disbands between different layers of the OPV and other panels.
- Advanced software algorithms to highlight the deformation anomalies, which in turn reveals surface or sub-surface defects;
- Comprehensive algorithms and AI-driven system will be developed for automatic defect recognition. These algorithms will build on fringe processing, edge detection, dimensional measurement, localisation of centroid of a target and pattern recognition methods based on statistical model and artificial neural network.

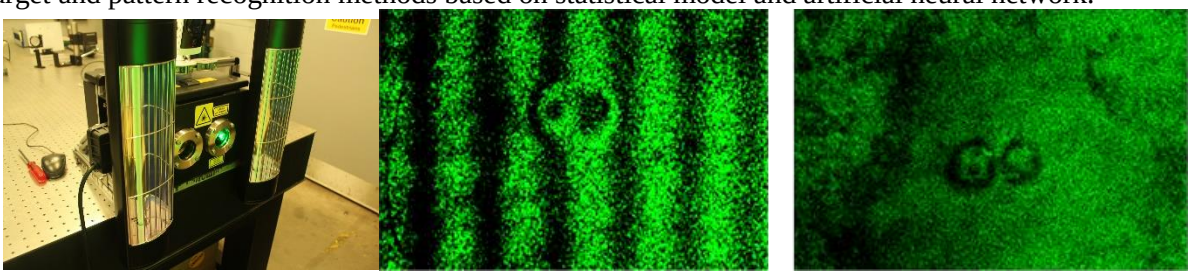


Figure 22 (a) Shearography system at TWI (b,c) fringe pattern showing subsurface defects (representative example)

OLED material screening using the robotic spin coating platform

Design of the experiment: number of materials, material combinations, layers in the stack, concentrations, solvent class. 2) Preparation of stock solution at concentrations calculated in the design, preparation of substrates, preparation of the platform, 3) Automated generation of code from the experimental design file, 4) Automated run of 49 samples. All first layers are completed before the next. Completed layers are temporarily stored in N2. Processing in parallel using two spin coaters. 5) Thermal evaporation of the top electrodes on 49 samples, 6) Semi-automated J-V-L characterization.

Figure 23 The robotic spin coating platform for combinatorial material testing.

Inspection is a crucial part of all manufacturing workflows at CSEM Muttenz. Since its manufacturing infrastructure consists of stand-alone S2S equipment, rather than a single manufacturing line, it is well suited for the development of new processes or process flows. The quality of the manufacturing process is controlled by visual inspection between each process step. The thicknesses of thin layers are characterized using profilometry or AFM; the shape of fabricated nano/microstructures using AFM or SEM; the quality of optical coatings by transmission spectroscopy. The topography over a larger area of micro structured surfaces is inspected by optical coherence tomography.

Automated Defect Recognition (ADR) system implemented with image processing and machine learning.

Technology development: An enhanced technique using image processing will be developed for determining the critical defects in the OLED/OPV component based on the acquired signal (i.e. images obtained by optical diffraction system), to ascertain the defect size and type to a high degree of certainty (>95 accuracy). Different filtering and thresholding algorithms can be evaluated and compared in term of their effectiveness on defect identification.
The machine learning technique will potentially use deep learning neural network to detect and classify defects.

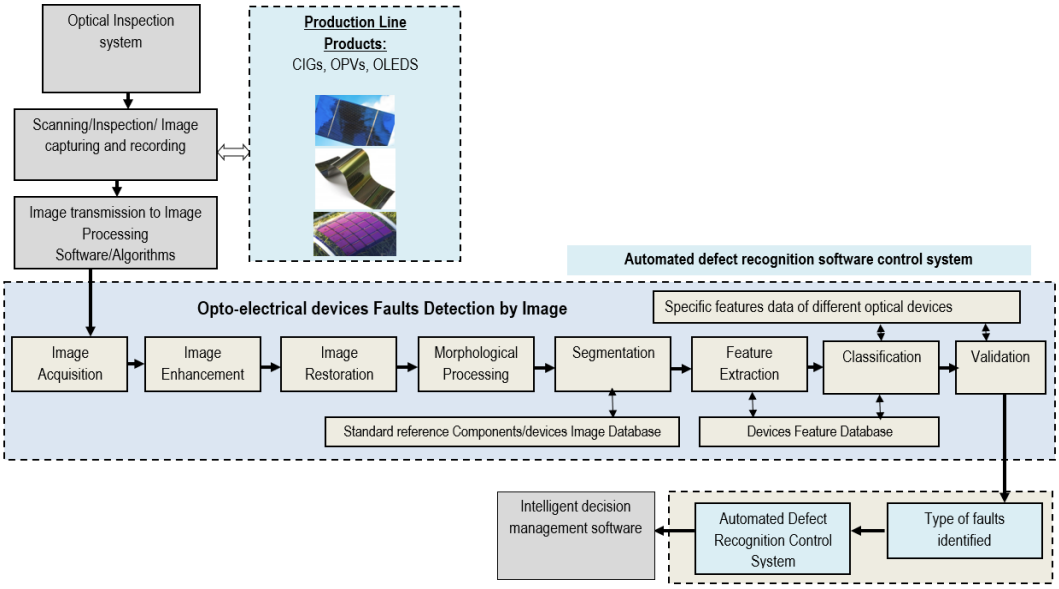


Figure 24 Data processing flow diagram

(1) Image processing

For part characterization based on X-ray, the required image processing steps have previously been investigated in image processing and computer vision[21]. A flow diagram is shown in Figure 1 for the image processing procedure for automatic inspection. There are four key stages: image restoration and enhancement, image registration and subtraction, image segmentation, and morphological operation. Each stage is described below in details.

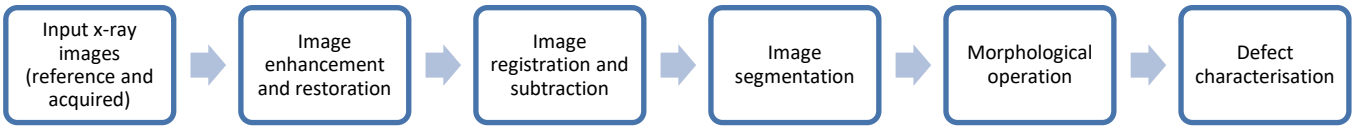


Figure 25 Typical ADR flow

The above workflow will be adopted for image processing in OLEDSOLAR. An example of the ADR process based on acquired x-ray images analysed and tested by BUL are illustrated in figures below.

Besides applying a single method for thresholding, hybrid methods can also be used for improved performance. Subsequently, on the basis of set theory, simple mathematical operators are derived for morphology, which can be exploited in this project to remove noise, enhance image, and detect object edges in the binary images. An example of such ADR process is applied successfully in acoustic signal to produce image is shown in Figure below. A characterization of object (i.e. defect) could be represented in Figure below which shows the comparison of unprocessed and image-processed (via the ADR process) results, in which the latter clearly demonstrates a significant boost of confidence in the results.

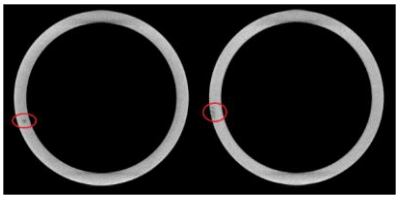


Figure 26 Original 2D image slices of the material under test with defects

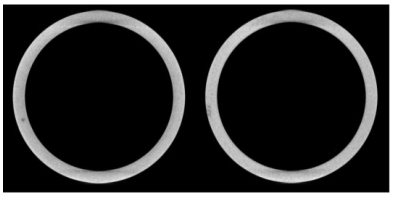


Figure 27 After denoising with Wiener filter

Figure 28 Before and after image registration

[21] A. Rosenfeld and A. C. Kak: 'Digital picture processing', 2nd edn, Vol. 2, 1982, New York, Academic.,
T. Y. Kong and A. Rosenfeld: 'Digital topology: introduction and survey', Comput. Vision Graph., 1989, 48, 357–393.J. K. Udupa: 'Multidimensional digital boundaries', CVGIP: Graph. Model Im. Proc., 1994, 56, 311–323.

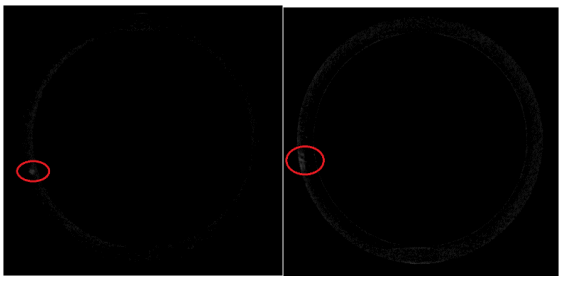


Figure 29 Difference images after image subtraction

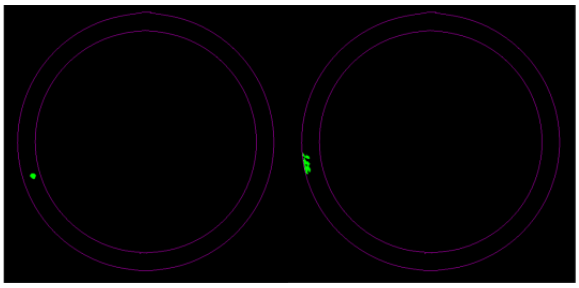


Figure 30 Thresholded images after morphological operation

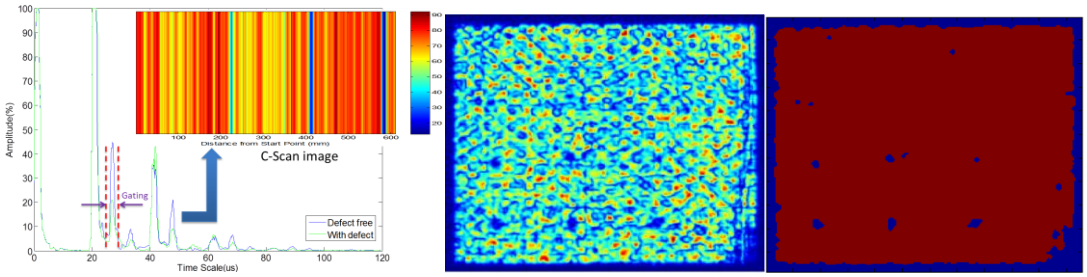


Figure 31 Example of ADR process applied to acoustic signal and (on elft) Comparison of unprocessed and processed image which the latter improves the confidence level of object identification in which the size can be estimated.

(a) Unprocessed image (b) Processed image to identify objects

(2) Machine Learning for defect detection and classification

Machine Learning has been shown to deliver state of the art results in data science applications. One of the great achievement in the machine learning technology during recent years is the Deep Learning, which are being successfully applied in a wide range of applications in business, medicine, geology and physics to solve problem of prediction, classification and control. Deep Learning network require fewer training samples and generate higher accuracy than the conventional back-propagation algorithm. With proper pre-process and feature extraction, the deep learning network can classify more complicated raw data and very abstract data. It can handle a variety of image classification tasks in industrial vision environments, such as fault detection, object recognition, operation prediction, engine controlling etc.

Advantage of Machine Learning Method for OLED/OPV Inspection:

- **Feasible**. Providing enough and well selected training strategy, the machine learning methods can be used to handle various complicated cases.
- **Robust**: Machine Learning methods are applicable in almost every complicated situation where a relationship between input and output parameters exists, even if the relationship is very complex and cannot be expressed or handled by other modelling methods.
- **Controllable**. By selecting proper training samples and network parameters, the machine learning method can be tailored to process various specific inspection task.
- **Self-Adaptive**: Neural Network is an adaptive system that changes its structure based on information provided in the training process. The machine learning strategy and parameters can be further adjusted based on the onsite accuracy.
- **Natural**. The basic principal of feature extraction, train sample selection, network training, and classifying is to simulate the human process, and results can be compared and reviewed by human inspection.

Data Acquisition: The proper selection of training data can affect the speed of training process, the chance of stuck in local minima, overall accuracy of recognition results, and tolerance to image quality. The network is expected to not only detect defect, but also group different type of defects, and must be able to process image under various quality. The training samples are selected from the data set of pre-processed x-ray images. A variety of faults will be simulated. The sample without any detected are collected as background images. The defected training samples are manually selected, classified and labelled. The training images are required to cover various type of defects, and various extent of noise and image quality. The defect training dataset consists of several subgroups, each having its own central tendency towards a particular pattern. The defect samples are required to represent every group of patterns. All of these patterns must be well represented. Within each subgroup, statistical variation must be adequately represented. The range of defect must represent every range of data including various situation of noise. In certain cases, only lower and upper bound data with few scattered data samples will be adequate if the variation can be considered as linear. The images free of defected are used as background data. Experiment shows that training with background data could dramatically reduce false recognition and improve the precision, even if the background training data are not exhaustive.

Data Augment and Normalization: As the OLED image could be any direction, the training data need to be transformed into various viewing positions, translations, reflections, and sizes. In the view of deep learning network, the effect of data augment is to avoid overfitting. Overfitting is a result that the training process failure to capture the genuine relationship between training samples and expected results. The overfitting network usually have large difference between the accuracy of classifying the training samples and accuracy of classifying the testing samples, especially when the network is in large scale.The common method of data augment is to extend the dataset using label-preserving transformation. One of the basic requirement of data transformation is to produce new data from the original image with very little computation, in order to avoid the transformed image to be swapped on disk which is very time consuming. For a large scale network and complicated dataset which require repeated experiments, the data swap on the disk should always been avoid as it could dramatically affect the progress of product development. The data augmentation will potentially be implemented in Python on the CPU, while the training on the previous batch of images will be processed on GPU.

The Network structures: Deep learning network have multi-layers, in which numerous similar elementary processing units (neurons) connected together each other between layers and constructed into a network. The processed data of each layer passes through the network towards the output layer.

The network for defect detecting is consist of multi-layers between input and output in the convolutional architectures, which called the Convolutional Neural Network. Such architecture allows richer intermediate representation to be created and trained. During the training process each layer is required to detect specific range of features at different level, a synthetic strategy which simulates the human brain learning process. Usually the front hidden layers are trained to detect lower level features like edges and lines, while later hidden layers are being trained to detect high level features. The multi-levels feature analysing method gives the deep learning network more powerful ability to extract useful intrinsic difference and relation between each classes, which will be used to classify objects based on the attributes of each object.

In general, large data set requires large scale network. However large network not always improve accuracy and sometimes it could result in overfitting. Large network also cost more training time. The size of network also depends on some other factors like the hardware, available memory, and batch size of training.

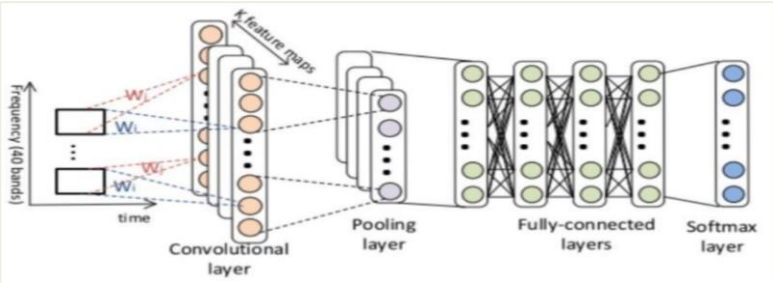


Figure 32 AI models

Some parameters need to be fine-tuned by experiments. The architecture of our network may contains 8 learned layers. Layer 1,2,3,4,5 are Convolutional layers. Layer 6,7,8 are fully connected layers.

Convolutional layers compute the output of neurons that are connected to local regions in the input, each computing a dot product between their weights and a small region they are connected to in the input volume. The kernels of the convolutional layer 2,4,5 are connected to those kernel maps in the previous layer which reside on the same GPU. The kernels of the convolutional layer 3 are connected to all kernel maps in the layer 2. The neurons in the fully-connected layers are connected to all neurons in the previous layer. Response-normalization layers follow the convolutional layers 1,2. Max-pooling layers follow both response-normalization layers as well as the fifth convolutional layer.

The ReLU (Rectified Linear Units) non-linearity layer is applied to the output of every convolutional and fully-connected layer. Experiment shows that the ReLU structure can improve the training speed several times for the convolutional neural networks. The other types of neuron activation includes: Sigmoid, Tangent/Hyperbolic, Absolute value, Power of shift and scale, logarithm of exponent etc. Each convolutional layer is followed by a Pooling layer, which is used to progressively reduce the spatial size to reduce the amount of computation in the network and also control the overfitting.

Layer 6,7,8 are implemented with 'dropout' parameter, which can roughly double the training speed. The basic idea of 'dropout' is to randomly set output of some hidden neuron to 0, usually with probability of 0.5. The dropout neurons do not contribute to the forward pass and do not participate in back-propagation. So every time an input is presented, the neural network samples a different architecture, but all these architectures share weights. This technique reduces complex co-adaptations of neurons. When a neuron cannot rely on the presence of particular other neurons, it will be forced to learn more robust features that are useful in conjunction with many different random subsets of the other neurons.

The output of the last fully-connected layer is fed to a multi-way Softmax sub-layer which produces a distribution over the all class labels. The network maximizes the multinomial logistic regression objective, which is equivalent

to maximizing the average across training cases of the log-probability of the correct label under the prediction distribution.

During classification, the network transform the original image layer by layer from the original pixel array to the final classifying results with confidence score. In each layer all the neurons multiply their outputs by 0.5, which is a reasonable approximation to taking the geometric mean of the predictive distributions produced by the exponentially-many dropout networks.

Training and Testing Procedures

During training cycles the network adapt the weights of neurons based on training sample which represents the desired outputs. After sufficient training, the network is able to relate the problem data to the appropriate solution data. The training image data will be manually labelled and cropped.

There are two type of adaptive learning method in deep learning:

a) Supervised where the network is provided with a large number of known inputs. Thereafter, the network produces the known outputs as closely as possible based on training.
b) Unsupervised learning, the network is required to work out relationships between various inputs without being told.

The first layer uses supervised learning method, while other hidden layers use unsupervised learning method.

The training and testing scripts will be implemented in Matlab and Python on Linux system.

The accuracy is measured by Recall Rate and Precision on the testing samples which are not presented during the training process.

By analysing the testing results the training samples probably need to be re-selected or re-generated in order to further improve the accuracy. However it's not necessary to repeat a new training from scratch. Experiment shows that a repeated new training can be dramatically accelerated by applying a previous trained model to the several front layers of network. That's because the front layers detect low level features which are very similar between different dataset.

Automation and Control in production line:

By developing OLEDSOLAR project, a novel control and monitoring system using PAT technologies will be applied whereby the industry brings the lab to the production process obtaining relevant data, in real time through dedicated sensors. These sensors can range from simple temperature probes to complex optical analysers which perform on-line chemical analysis or deposition layers. With this information and after applying deep-learning and process control techniques we will be able to provide the operator with the essential information to efficiently improve their process in terms of detection of film thickness, yield and other coating features.

IRIS has tested in previous projects a monitoring system called PATBox, PATBox its monitoring system that has the following implementation stages of PAT systems during the production processes for OLED and solar cells:

1) Data collection is made by using analytical probes located at critical control points of different processes industries of the optoelectronic devices.
2) To develop stochastic-data-based predictive models by applying data mining tools to the analytical data.
3) To optimize the process according to the best model achieved by collecting and mining data. 4)To perform model maintenance along the life-cycle of the optimized production line.

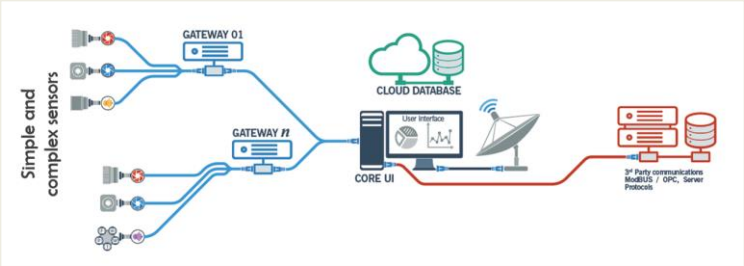


Figure 33 Scheme of the system architecture

PATBox will become a versatile global control system for OLEDSOLAR that will acquire and processes the data for closed loop control of the process of for decision support.

3. Recycling and Re-use Process development:

IN OLEDSOLAR, an innovative approach will be used for the evaluation and selection of main techniques (procedures and technologies) for the optimal reuse and/or recycling of OLED and solar panels components (as wastes) both during manufacturing process as well as at the end of their life cycle. In detail, the approach will include the identification, quantification and analysis of wastes produced during the manufacturing process by using Key Performance Indicators (KPIs), the analysis of their potential reusability into the production line according to their related costs (raw material and/or production cost), definition of the potential handling and separation and finally, the feasibility evaluation of the following recycling routes: ***(a)*** Grinding/homogenisation for external recycling by using existing technologies (when needed), ***(b)*** manual sorting and pre-concentration of added value materials before external recycling, and ***(c)*** internal recycling by using novel solutions.

Particularly for the reusability potential, the reuse of the OLEDSOLAR manufacturing wastes will be evaluated in terms of potential loose of material properties. Regarding the validation of recycling technologies, mature and novel pre-treatment technologies will be tested and evaluated at laboratory and/or pilot scale, depending on laboratory results. Mechanical (i.e. grinding), thermal (i.e. heating) and chemical (i.e. leaching) technologies will be considered.

The project conceptual approach is depicted in figure below:

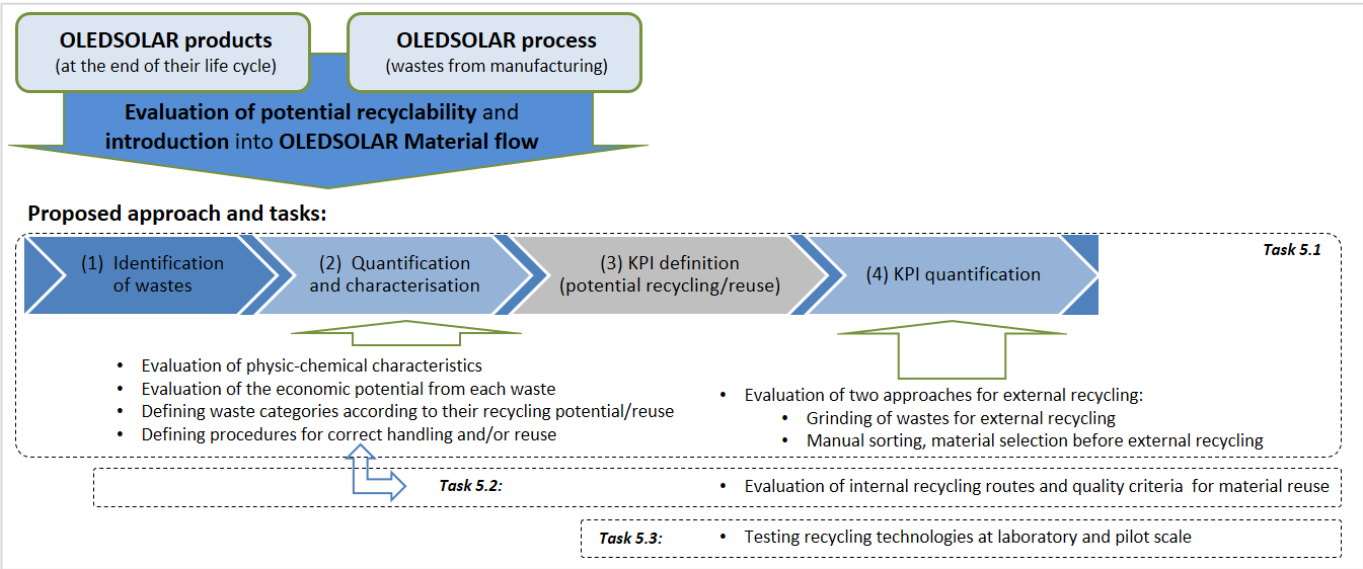


Figure 34 Recycling conceptual approach

Pilot Line and Production facilities for testing and demonstration of OLEDSOLAR solution:
VTT will offer pilot line facilities for testing the OLEDSOLAR concept.
The most important OLEDSOLAR-related pilot lines are the following:

- **MAXI printing line**
  - 4 interchangeable printing unit slots
  - Available deposition units: forward and reverse gravure, rotary silk screen,
  flexography and slot die coating
  - Plasma substrate treatment unit
  - Lamination unit
  - R2R hot embossing unit
  - Die cutting unit
  - Drying units (air, UV)
  - Automatic registration system
  - Max. web width 300mm
  - Max. web velocity 30m/min

Figure 35 VTT's Pilot Line (Printing Equipment)

DELTA converting line

- 6 interchangeable converting unit slots
  - kiss cut (top and bottom)
  - lamination
  - slitting

Figure 36 VTT's pilot line

- CO2 laser
- Automatic registration system for lamination
- Max. web width 300mm
- Max. web velocity 90m/min

Figure 37 VTT's pilot line (testing equipment)

TESLA2 R2R testing machine

- 192 measurement channels
- Minimum test pad size 2mm x 2mm
- Maximum test fixture size 408mm x 290mm
- Automated test fixture alignment
- In-house test fixture design and assembly
- Different measurement possibilities between test points
    - LCR measurements
    - DC Power supplies (0-60V, 0-8A)
    - Electrical load
    - OPV characterization (IV-curve)
    - OLED and LED functionality testing

The pilot facilities will be used to develop automatic registration of transparent/non-transparent inks and high-resolution imaging of printing quality. The precision requirements of traditional printing come from the resolution of the human eye. This accuracy is considerably worse than the precision requirements of the printed electronics, where desire is often to achieve micrometre precision. Therefore, the registration cameras used in traditional printing systems are not suitable for printed electronics. It's not just about the need for greater magnification because the fast-moving line combined with the micrometric resolution needs also a camera with very short exposure time. If, for example, the speed is relatively slow, i.e. only 10m / min (= 0.167m / s), and it is desired that the line movement does not exceed 1µm during shooting, the maximum allowable exposure time is only 1µm / 0.167 m / s = 6 µs.

Traditionally, this has demanded costly special arrangements for both cameras and lighting. However, VTT has experience in camera systems with a low-cost illumination solution based on pulsed high power led. This illumination / camera solution will also be used in the camera build in this project. Figure X shows examples of three different cameras that have been utilised at VTT to optimise the registration. The image below the cameras show how the registration accuracy has been improved from ±150 µm to ± 50 µm. In the OLEDSOLAR project, these methods will be developed further and finally transferred to the industrial production lines.

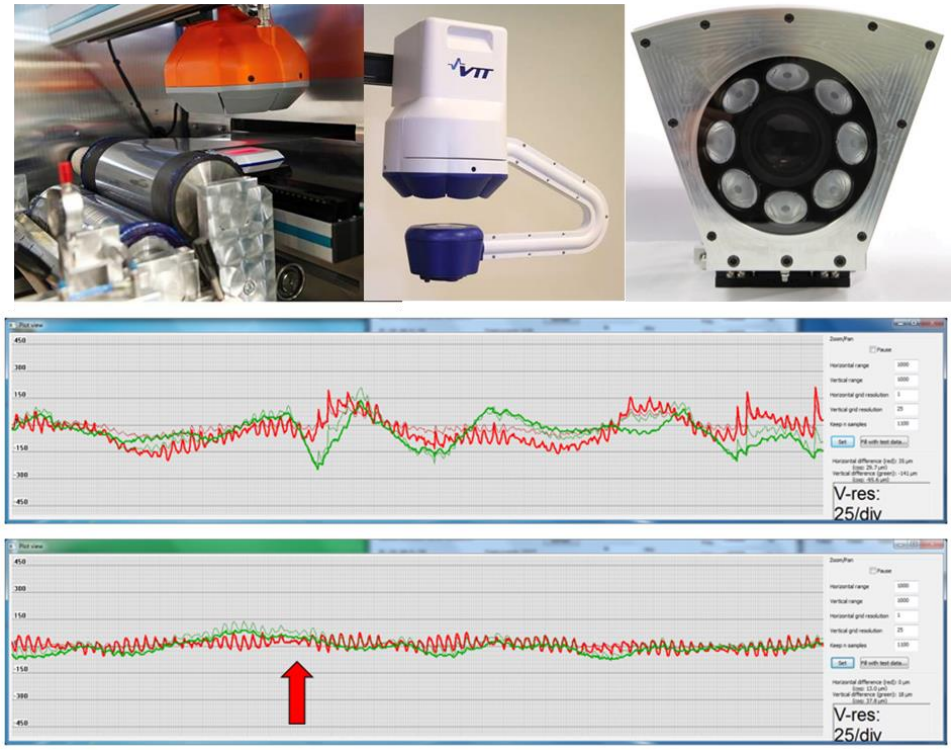
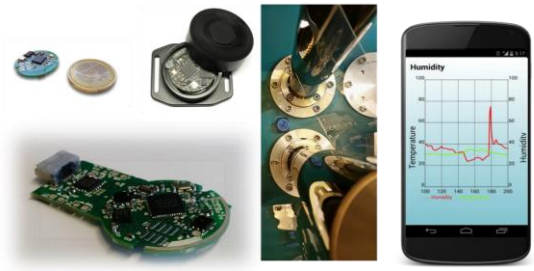


Figure 38 Three different cameras, which have been developed at VTT (top) and the resulting improvement in the registration accuracy (bottom).

In addition, the existing VTT's TinyNode wireless sensor system (accelerometer/3D, pressure, inclinometer/3D, magnetometer/3D, humidity, temperature, particles, web tension etc.) at MAXI can be used to collect data during processing, and new sensors based on requirements from industrial partners can be added to the



system. Figure Y shows examples of the sensors and an image of the mobile phone app for following the temperature and humidity measurements.

Figure 39 Pictures of VTT's TinyNode sensors (left), their installation at the MAXI line (middle) and an example of the mobile phone-based application to follow humidity and temperature (right).

Figure Y. Pictures of VTT's TinyNode sensors (left), their installation at the MAXI line (middle) and an example of the mobile phone-based application to follow humidity and temperature (right).
The data from the R2R functional tester, imaging and sensors will be used as material for developing software and algorithms for machine learning -based methods (by BUL) to correlate the production conditions, printing quality and electrical performance.

Case Study: Testing CIGs product manufacturing in production line of Solibro Hi Tech:

Solibro-Hi Tech Production facility: The developed solution of OLEDSOLAR will be tested at production pilot line of Solibro Hi Tech in CIGs manufacturing.

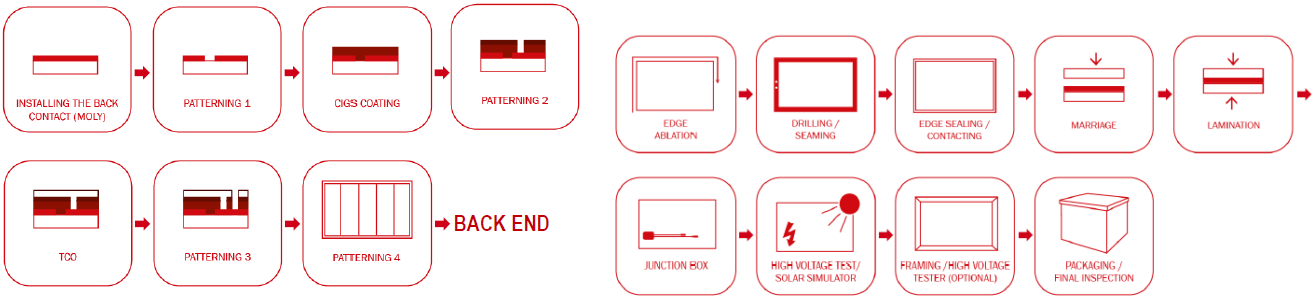


Figure 40 Solibro Production line and front end/back end process

Problem and Metrology Testing requirements of Solibro

- With the current metrology of the CIGS deposition there is limited inline access to the optical & electrical quality of the absorber layer. To improve the yield, process window and to get a faster feedback during the introduction of new processes & materials a new inline measurement system is envisaged. The use of spectral and/or transient photoluminescence (PL) has shown promise on laboratory scale to supply valuable information about the recombination properties of the absorber and its interfaces enabling a forward calculation of the Voc. Therefore, in OLEDSOLAR it is proposed to develop an inline PL system in order to investigate the correlation of the absorber quality with its properties under field conditions. The aim is to evaluate and install a method to characterize the CIGS layer in line with high sampling rate and local resolution. The optimum PL characterization method and position within the process chain should be identified, which enables the best prediction of the field properties based on inline measurement.
- The incorporation of a new Boron doped Zinc oxide (BZO) into the CIGS solar cell stack creates difficulties compared to the traditional Aluminum doped Zinc oxide (AZO). The transparent BZO layer grows with large grains that create a very rough surface. While single layers on glass can be characterized at reasonable expenses determining layer properties inline / on CIGS such as thickness, sheet resistance, haze or crystallinity is very challenging. Due to the nature of the layer stack and the combination with the rough surface texture new concepts need to be evaluated in order to enable an inline process control for the LPCVD BZO process. The systems in question needs to access the material properties within a reasonable amount of time (ideally a few seconds; one minute at maximum), operate non-destructively and has to be flexible in terms of the substrate position as the glass typically exhibits a warp of a few millimetres. *The proposed inspection system (detailed in WP3) and NDT techniques will enable fast and accurate inspection of proposed materials in CIGS manufacturing with high level of accuracy.*

Case Study: Testing OPVs product manufacturing in production line of ASPF:
ASPF OPV manufacturing process

The materials used for the fabrication of OPVs are predominantly carbon based with no rare or toxic elements included. These organic compounds act as building blocks for the solar cells and can be easily and cost effectively synthesized in large quantities and high quality. The chemical and physical properties of the organic materials and therefore the OPV characteristics can be furthermore tailored to the customer's demands and expectations. This assures a wide range of application possibilities for the OPVs, due to their adjustable colour, shape, and viable power output.

The organic materials used for the manufacturing processes of the OPVs are solubilized. Their deposition, therefore, can be accomplished by means of common printing techniques, avoiding expensive vacuum based techniques and high temperatures. Similar to the printing of newspapers and magazines, OPVs can be fabricated on thin plastics at a high throughput on a continuously moving roll-to-roll web. The extremely low energy consumption during the manufacturing process compared to their inorganic silicon counterparts, leads to a very low carbon footprint. At the end of their life, some of the materials within the OPV structure can be recycled and reused, while the OPV laminates can be disposed. These factors show that the OPV technology allows a material, energy, and carbon savings from "cradle to grave" and represents a truly green renewable energy solution.

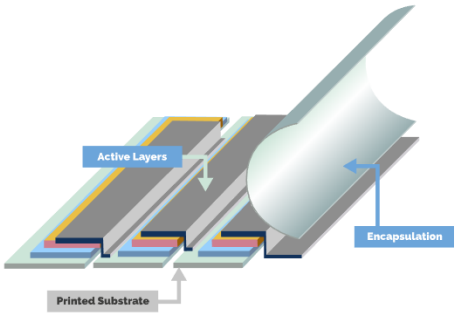


Figure 41 OPVs layer structure

Process steps

Since organic photovoltaics can be integrated into nearly every surface and product, due to their mechanical flexibly, low weight, and possible semi-transparency, the manufacturing process will be adapted to the customer's requirements. Nevertheless, these steps are important:

Figure 42 ASPF MANUFACTURING PROCESS

As shown above, firstly the substrate carrier has to be prepared for the sequential deposition of the OPV layers. For this step, lasers are used to structure the bottom transparent electrode on top of the substrate, allowing the creation of OPV base with numberless shapes, some of which are shown above. Then, the layers of the OPV are deposited using printing techniques, allowing easy scalability of the solar cell' size. OPVs with different colours are also possible. The top electrode deposition follows and is required to complete the OPV device. The design of the top electrode can be chosen by the customer and can represent a particular figure or a logo. At the end, the OPV is encapsulated to assure long term operational stability.

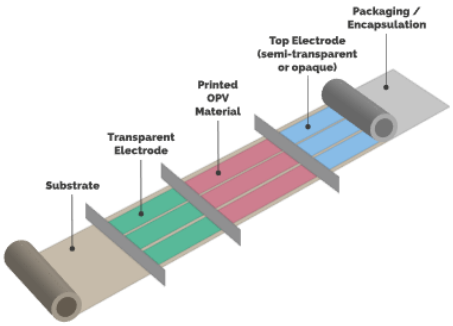


Figure 43 OPVs processing steps at ASPF

In OLEDSOLAR project deposition technique, patterning processes and packaging processes will be implemented and tested in ASPF production line. Inspection system and sensors will be installed for testing and performance validation as well. In summary following validations will be done in OPVs production of ASPF:

- Automation of registration of screen-printing for free-form OPV modules including transparent inks;
- Analysis of coated layers of OPVs in a way to predict a performance of an unstructured area. The following methods proposed in OLEDSOLAR will be implemented for analysing the coated layers could be:
  - Image analyse (VIS, evtl. IR) for dust, scratched, defects;
  - Thickness measurement;
  - Photoluminescence-scanning (x,y);
  - ILIT (illuminated lock-in thermography);
- After testing the above methods, together with further processing data (temperature at the various coating and/or drying units, speed, …) such data will be further processed using Big-Data/Machine-Learning-Algorithms developed in OLEDSOLAR by BUL. This could on one hand lead to a further reduction of waste and support on the other hand the development for the prediction of performance of unstructured coatings in OPVs production of ASPF.

Case Study: Testing thin films enabling opto-electronic devices manufacturing (DuPont Teijin Film)

OLEDSOLAR solution will be implemented and tested in production line of DuPont Teijin film to validate and improve manufacturing of thin films one of the main component in OLEDs and solar cells efficiency,

- Assessing suitability of existing equipment suitable for upgrade as per OLEDSOLAR inspection system adaptability
- Qualification, testing and validation of barrier films performance
- Implementing OLEDSOLAR inspection and testing systems to analyse thin film performance for surface quality, specification validation and performance verification
- Automated defect detection and processing during thin film production,
- Assessing recycling and re-use strategies being developed in OLEDSOLAR within production of thin films

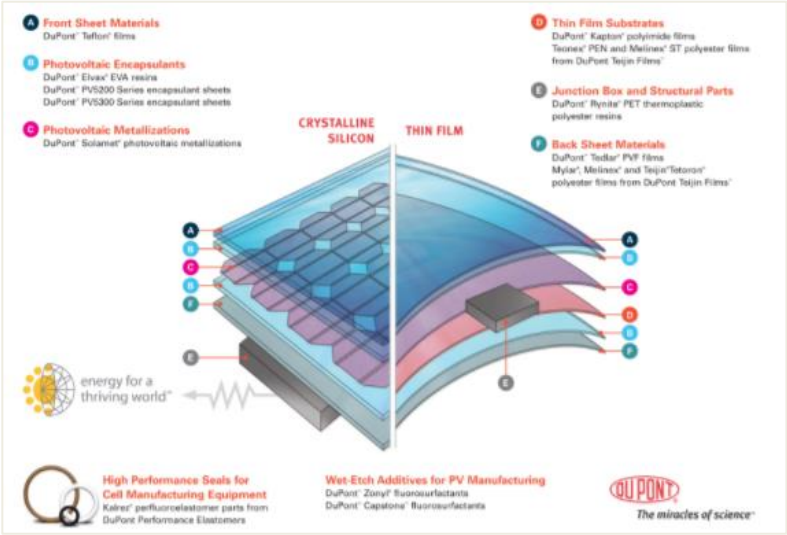


Figure 44 Thin film for solar cells

Project Positioning:

The project positioning along with TRL is depicted in figure below:

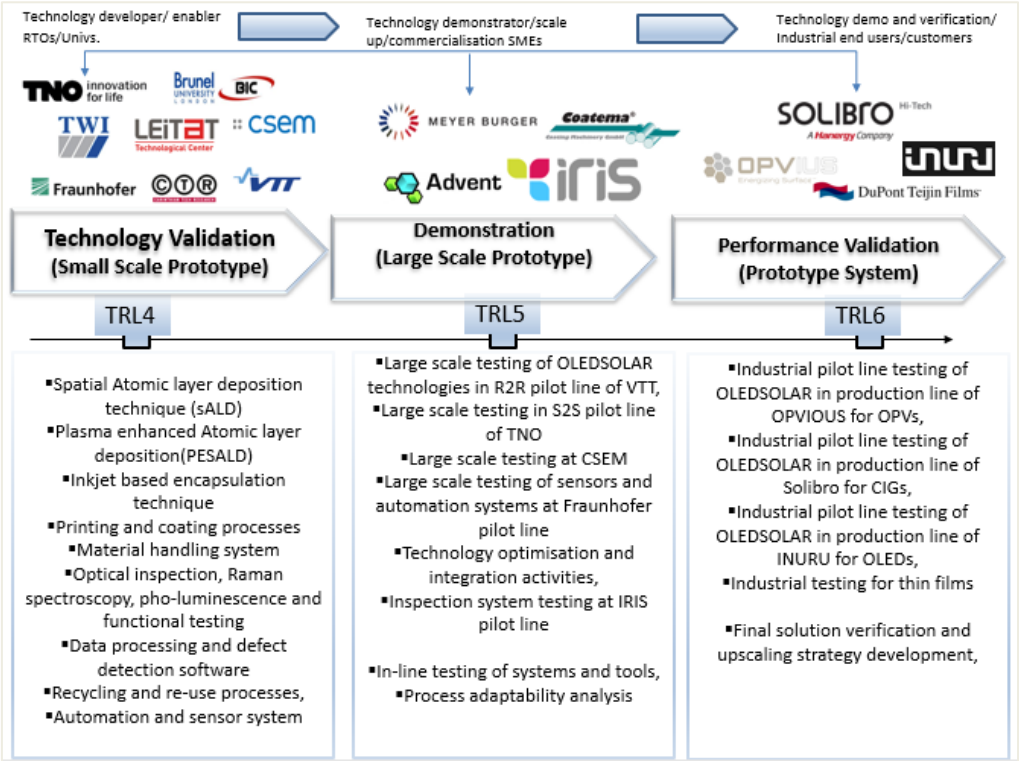


Figure 45 Project positioning and TRL

Some of the linked international/national activities of partners feeding input into the project are listed in table below:

Table 2- Interlinked national and international activities and feeding results into OLEDSOLAR

National and International Research and Innovation Activities	Partner	Relevant Results feeding inputs to OLEDSOLAR

OPTINANOPRO (H2020, GA Nr: 686116)	IRIS	Development to TRL5 of optical methods for inline real-time thickness monitoring of coatings and lamination, compounding, injection/co-injection and electrospray Nano deposition.
FP7 THIME		Results related to thin film measurements on organic photovoltaic layers,
FLEXOLIGHTING	BUL	Results of the demonstrator being built in the project for OLED lightning application,
AutoDISC		Image processing
OLEDSOLAR		Image processing
Qualinet		Image processing
Assurenet		Image processing
iFROG		Image processing
FobFMS		Image processing & machine learning
Intelliscan		Image processing & machine learning
RobotX		Image processing & machine learning
Austrian COMET compentence centre programme – K1 centre ASSIC (www.assic.eu)	CTR	Material analysis & characterization of semiconductor and polymer materials using Raman spectroscopy (lab + process)
Optima		In-line contamination and quality control of high-performance polymers and polymer parts for semiconductor manufacturing equipment
EVAnetz		In-line quality control (degree of crosslinking) of embedding materials (E/VA) for industrial PV module manufacturing (incl. tools for process development, optimisation and validation)
Infinity		Development of climate-zone-sensitive high-performance PV systems along the full value chain: material – components – modules – module-integrated electronics – inverters – system integration – control & maintenance strategy – in-field inspection tools (EL, PL, lock-in thermography, electrical …)
Pi-Scale (GA: 688093); PhotonFlex (EFRE: 0800069); HEA2D (EFRE: 0800148); INTRES (EFRE: 0800645); GreenSense (GA: 761000)	Coatema Coating Machinery	Intermittent slot-die coating for pilot lines; coating & process development for OPV; coating of transfer polymer & transfer of graphene; coating and process development for batteries; coating of foil substrates and free-standing Nanocellulose;
Alabo	FRAUNHOFER	Evaluation of ultra-barrier webs
Printed electronics (OPV, OLED, OTFT), material validation, device design and integration, prototyping	CSEM	Knowhow on device fabrication, providing S2S fabrication infrastructure, environment for prototyping and testing, dedicated methods for testing such as optical Ca test
THIME, Pi-Scale, ArtESun, Monoscribe, IN-LIGHT, Polaric etc.	VTT	R2R and S2S process and characterization knowhow for OPVs, CIGSs, OLEDs and OTFTs, sensor development for pilot lines, inline quality control.
Detailed list of projects and partners involved is detailed in section 4		

Standards, Certifications, Safety codes: OLEDSOLAR project will address following standards.
Standard in the series, IEC international standard 62341-6-1:2017, which "specifies the standard measurement conditions and measuring methods for determining optical and electro-optical parameters of OLED display modules and, where specified, OLED display panels.LEITAT and TWI will work to make sure compliance with standards.

(b) Methodology

The project proposed methodology is depicted in flow diagram below:

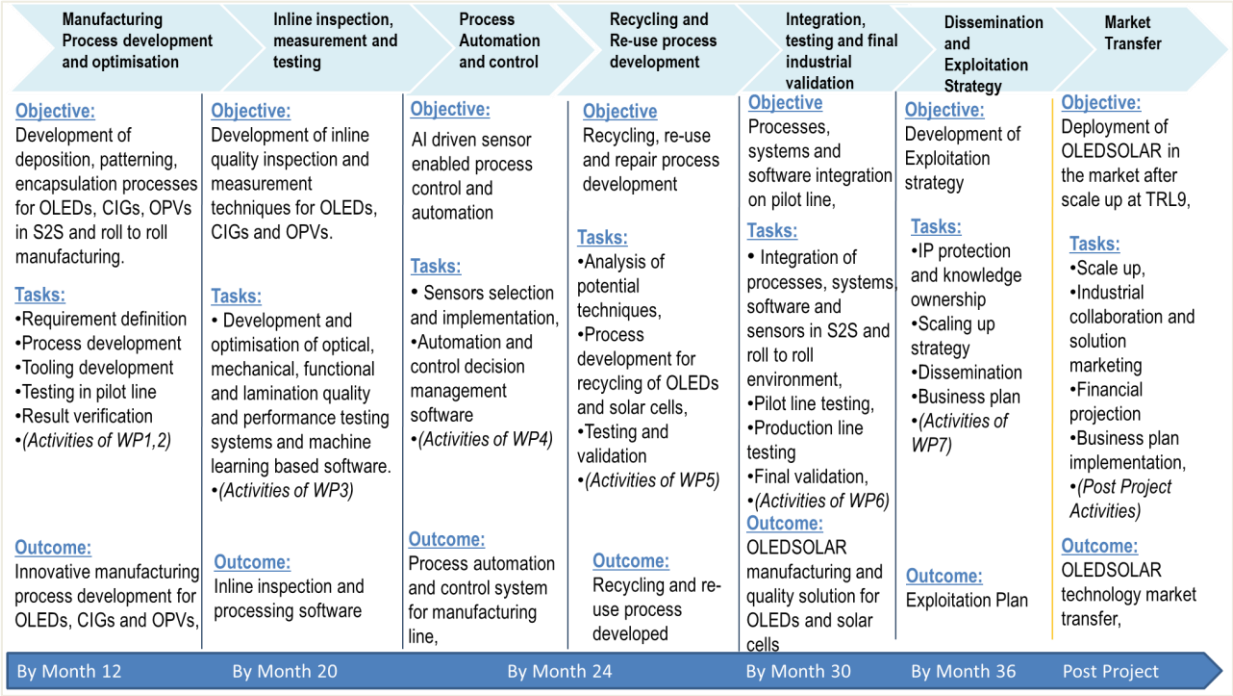


Figure 46 Project methodology

Gender Analysis: OLEDSOLAR will develop and implement measuresto eradicate gender-based disparities and promote gender equality in accordance with the EU Strategy for equality between women and men[22].

1.4 Ambition

The alternative approaches and limitations vs. OLEDSOLAR competitive advantage is detailed in table below:

Table 3: Competitive comparison of OLEDSOLAR with other techniques for targeted devices

Devices issues	Limitations of alternate devices and processes
OLED: OLED suppliers claim the technology offers far more colours, brightness and contrast with less motion blur than LCD, especially against cheaper LCD models	LCD/Plasma: Limits in brilliance, resolution and colour ranges due to their inherent structures and physical processes of displaying images.
OLED: Power and luminescent efficiency are higher as OLEDs do not use light from a backplane backlight (the transmissive mode) but from the polymer (emissive mode).	LCD/Plasma: Power requirements that are difficult to reduce further, owing to the need for a backlight for LCD, despite the move to large LEDs, or physical discharge for plasma
OLED: Inherent physical properties of polymers enable flexible displays to be made and this widens the applications base enormously.	LCD/Plasma: The mainstream technologies in LCD and plasma do not produce flexible types of displays.
OLED: The display's weight can be lower as there is no backlight and the whole unit can also be made thinner, of the order of a few millimetres thick.	LCD/Plasma: Weights that are difficult to reduce further because of the need for complex substrates and backlighting planes.
OPV: Emerging photovoltaic cells are made from an assortment of inexpensive materials, including organic polymers, small molecules, and various types of inorganic compounds, which can be fabricated on flexible supports via inexpensive solution-phase techniques such as high-speed roll-to-roll printing.	Silicon solar cell: Because silicon does not absorb sunlight strongly, silicon cells contain a relatively thick layer of silicon, which is brittle and therefore must be supported on a rigid, heavy piece of glass, adding cost and limiting applications.
OPV: Low-temperature manufacture enables plastic or virtually any smooth support layer to be used	Silicon solar cell: High temperature during fabrication requires metal or glass support layer.

[22] COM(2010) 491 final.

OPV: Solar cells are lightweight, flexible and rugged	Silicon solar cell: Solar cells are heavy, rigid and fragile
OPV: Solar cells can be any color and semi-transparent	Silicon solar cell: Solar cells are black or blue
Thin-film encapsulation (CVD): We will be looking at integrate Atomic Layer Deposition process into R2R CIGS photovoltaics production sequence for complete protection of cells against humidity;	-CVD has no inherent control of the film thickness using surface chemistry -CVD can reach a limited degree of control only by extremely careful control of reaction time, reactant exposure and reaction temperature -CVD does not deposit conformally; rather deposition is determined by reactant flux -Reactants may react in the gas phase during CVD; consequently, particles may deposit on the surface resulting in a granular, nonconformal film

OLESOLAR Solution benefits	For roll-to-roll production of these components, the correlation between all parameters (materials, production, inspection) and the final device performance needs to be determined as early as possible. For instance, the performance of organic solar cells and organic LEDs is highly dependent on the thicknesses of the constituent layer materials composing the device. The lack of accurate information about the thickness of the thin films would prevent the detection of components that not perform well against design specifications. Without real-time thin film thickness measurements, or continuous yield monitoring directly integrated in the production line, the capacity of optimizing the production process becomes very limited and costly.
Benefits to industry	OLED – Industrial application (automotive) (opto-electrical part): Energy savings/Fuel efficiency: As OLED light uses less power compared to LEDs and other lighting solutions, the low-power usage will be the most attractive feature to car-makers and owners. Moreover, the low-weight OLED light panels will bring energy savings and increase fuel efficiency. Design flexibility: OLED light panels are also non-glaring, which means reflectors, light guides or similar optical components are not required. A thin (0.8mm - 1.5mm) OLED lighting strip stretched across any of the aerodynamically optimised surfaces could open a number of exciting new options. This will expand the design possibilities of car lights and provide design freedom to designers. **Luminance** quality: The uniformity of light output from an OLED panel is also advantageous as it will reduce the need for reflectors, which are necessary with other lighting techniques. Ultimately, automotive OLED lights printed on transparent plastics could also be fitted to the windows of cars too. On the other side, OLED light offers totally uniform, smooth and visually comfortable light with low-glare and low-shadow features. These characteristics of OLED light reduce eye fatigue and let people comfortably enjoy the light; **OPV** (opto-electrical part): Thin film photovoltaic cells based on solution processable organic semiconductors have attracted remarkable interest as a possible alternative to conventional, inorganic photovoltaic technologies. The following key advantages of organic photovoltaic (OPV) devices have been identified: Low weight and flexibility of the PV modules; Semitransparency; easy integration into other products; new market opportunities, e.g. wearable PV; significantly lower manufacturing costs compared to conventional inorganic technologies; Manufacturing of OPV in a continuous process using state of the art printing tools; Short energy payback times and low environmental impact during manufacturing and operations[23].

State of the art spatial ALD

[23] Efficiency of bulk-heterojunction organic solar cells

The first example of application of spatial ALD process has been the adoption of an Al_2O_3 layer as backside passivation film in crystalline silicon solar cell. Two companies, Levitech and SoLayTec focus on this technology to meet the industrial requirement of processing 3000 wafers/h, are currently selling commercial spatial ALD tools.[1–3] Flexible electronics such as light emitting diodes (OLED) as well as thin film photovoltaics are moving from the lab scale towards the production, and several challenges are still to be tackled. Most of the materials used in these devices are sensitive to O_2 and H_2O, and therefore encapsulation strategies which are economically viable, compatible with high-throughput and low cost deposition methods on large area are strictly required. For OLEDs, water vapor transmission rates (WVTRs) as low as 10^{-6} $g \cdot m^{-2} \cdot day^{-1}$are typically needed. For thin film photo-voltaic modules the WVTR is expected to be less stringent (depending on type of by one or two orders of magnitude) but still far beyond the simple polymer encapsulation foils (1 $g \cdot m^{-2} \cdot day^{-1}$). Conventional solutions such as glass-glass encapsulation are not compatible with flexible PV applications, and suffer from relative high weight. ALD layers such as Al_2O_3, multilayer stacks of Al_2O_3/TiO_2, combination of inorganic/organic layers have been already proven in lab scale as valid solution with excellent WVTR values.[4–10] Spatial ALD approach combined with a roll to roll (R2R) line can potentially meet the requirement of high throughput low cost barrier/encapsulation.

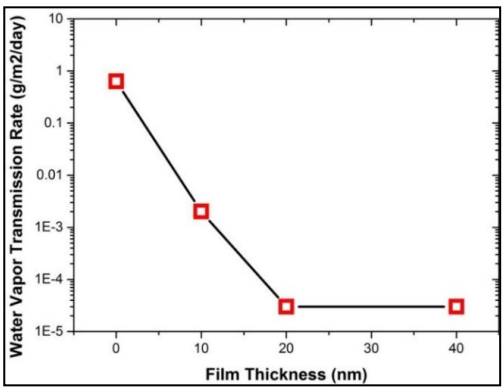


Figure 47 Water vapour transmission rate (WVTR) as a function of Al2O3 film deposited via roll to roll atmospheric pressure spatial ALD on PET foil.

Spatial ALD Al_2O_3 50 nm thick film has been deposited on glass substrates of an industrially relevant size (370 × 470 mm^2). The film exhibits a WVTR below the detection limit of the MOCON test (**3 x 10^{-3} $g \cdot m^{-2} \cdot day^{-1}$ at 20°C/50 %RH).**[11] Spatial ALD Al_2O_3 and SnO_2 have been demonstrated on a lab scale test as valid gas permeation barrier using oxygen plasma (PE-s-ALD) as oxidative source instead of the conventional water vapour. A 100nm thick Al_2O_3 layer deposited at 150 °C exhibits a WVTR of **5 x 10^{-5} $g \cdot m^{-2} \cdot day^{-1}$ (at 60°C/60 %RH**, and of **2 x 10^{-4} $g \cdot m^{-2} \cdot day^{-1}$ (at 60°C/60 %RH)** when the deposited at 100C.[12] In the case of 110 nm SnO_2 a value of **7 x 10^{-4} $g \cdot m^{-2} \cdot day^{-1}$ (at 60°C/60 %RH)** has been measured for the SnO_2 deposited at 150 °C, but one order of magnitude in WVTR has been achieved using a lower temperature (100 °C).[13]

TNO already tested spatial ALD layers on the own R2R pilot line. Depositing only 20 nm of Al_2O_3 by roll to roll atmospheric pressure spatial ALD on PET foil (**30cm x 2m**) without any pretreatment and/or planarization layer, drastically improves the WVTR value up to **2 x 10^{-5} $g \cdot m^{-2} \cdot day^{-1}$ (at 20°C/50 %RH) and 4 x 10^{-4} $g \cdot m^{-2} \cdot day^{-1}$ (at 60°C/90 %RH)**. Still improvements can be achieved developing "*ad hoc*" solutions/strategies.

Al_2O_3 layer has been successfully deposited also in the large area S2S-ALD tool, with an impressive uniformity (**less than 0.5% relative thickness**) over the entire glass plate (deposition area of 300 x 300 mm).

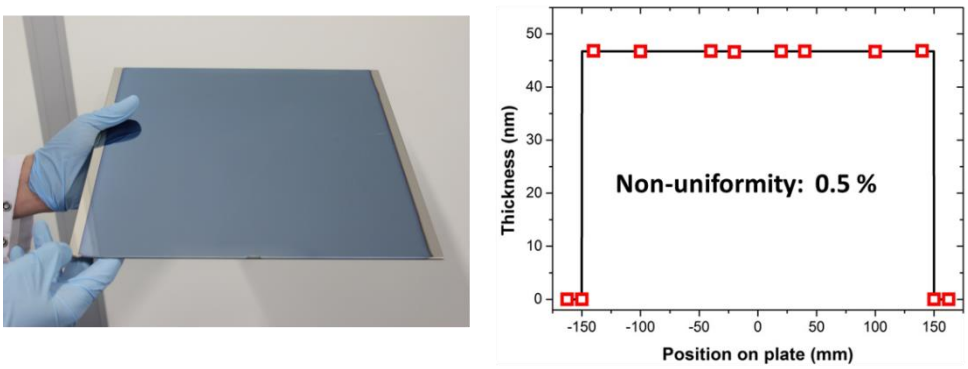


Figure 48 Example of Al2O3 deposited on large area S2S s-ALD reactor tool (on the left) and the thickness uniformity over the glass plate (on the right).

Furthermore, the same Al_2O_3 layer has been adopted as inorganic layer in the thin film encapsulation approach on flexible PET foil. This strategy, has been developed using plasma enhanced chemical vapour deposited (PECVD) SiNx layer in combination with organic printed layers. The thin film encapsulation which include large area Al_2O_3 spatial ALD layer in alternative to the PECVD SiNx, exhibited **no visible degradation after more than 1100 hrs of test at 60 ºC / 90 % RH**, which corresponds to an equivalent time of 2.5 years at ambient conditions.

Next to the barrier layers required for the OLED and solar cells, in the field of thin film photovoltaics, spatial ALD technique can be adopted to deposit functional layers in the device architectures. TNO developed a low temperature zinc oxysulfide (Zn (O,S)) process which has been used to fabricate an efficient buffer layer in cadmium-free CIGS device. By co-injecting the vaporized H_2O and H_2S gas in the same deposition zone a **low temperature (120 °C) buffer layer has been implemented on the CIGS stack**, exhibiting similar performance with respect to CIGS device

including the traditional chemical bath deposited CdS. The result proves the viability of spatial ALD as a new manufacturing technique to deposit and engineer functional layers in thin film solar cells. The low temperature enabled the application of the Zn(O,S) buffer layer on the roll-to-roll spatial ALD tool for flexible CIGS solar cells.

Progress beyond the state of the art

The results with lab-scale tools and large area deposition prototypes demonstrate that atmospheric spatial ALD layer is a promising technique to deposit barrier layers, as well as functional films for PV devices. However, several aspects in the parameters processing requires further investigations, improvements and other solutions can be explored. At the same time, it is necessary to evaluate the deposition yield, potentially in situ diagnostic, control of the coating, and the particle number which is detrimental to obtain pinhole free films.

It is necessary to identify the lowest requirements in terms of temperature and speed deposition in correlation with the barrier properties in order to upscale the process for mass production. While reducing the thermal budget of the s-ALD processes generally lower the quality barrier properties, alternative approaches to the single Al_2O_3 film such as nanolaminate, precursor mixing, introduction of organic layer (MLD), as well as use of atmospheric plasma technology might be a viable route to provide even better performance in terms of WVTR barrier properties. Being the barrier applied for flexible and potentially even rolled applications, the barrier needs to withstand mechanical stress without failures.

On the R2R-ALD prototype tool at TNO, barrier deposition is without plasma enhancement. Potential steps in OLEDSOLAR to enlarge capabilities for this or the envisaged bench (at Smit Thermal Solutions) are, to achieve in cooperation with metrology partners:

- Enlarge range of coating options beyond current one of Al_2O_3 and TiO_2
- Development of R2R-ALD test bench with plasma enhanced deposition
- Particle controls
- Integration with pre/post processing (e.g. application of planarization/protection polymer coatings)
- In-line inspection of coatings such as ellipsometry, Raman, PL for automatic adjustment of process settings (for enhanced process control)
- Re-cycling of gasses, exhausts, excess of precursors "

In-line inspection process state of the art and progress beyond state of the art:

State of the art:

A number of monitoring techniques exist when it comes to characterizing structures during the inspection and quality control in R2R environment for organic solar cells & OLED displays; in the nano- and micro-metre ranges, which have been utilised in the development of the LIPSS fabrication technique, including SEM (scanning electron microscopy) and AFM (atomic force microscopy) [24,25.] Fourier transform analysis of the SEM images have been used to quantify the periodicity of the structures. This approach enables a global value of spatial period to be quickly calculated, rather than estimating line thickness and spacings line by line, and generating a mean value, which is a more laborious process.

However, most commonly used techniques at this dimensional scale are in fact implemented off line and require sample preparation, Optical techniques offer the best opportunity for inline process monitoring, as they are non-contact, non-destructive and can be applied to samples as they are without pre-treatment. According to a recent review paper by IRIS[26], selected combined optical technologies, such as diffractometry, but also ellipsometry and spectral reflectance provide the opportunity for inline measurement of the thicknesses of nanostructured layers.

White light Interferometry: An inline spectral reflectance system was developed in the THIME project[27] to measure Organic PV films. White light interferometry has been developed into an inline system by Dr Schwab[28] for measuring the thickness of spacer and cover layers of Blu-Ray discs, but can also be applied to measure the surface topography of reflective materials. However, while this system measures topography with a high level of accuracy (<100nm), it's lateral resolution is limited to 10s of micrometres. An inline system has been developed by Nanometrics[29] employing spectral reflectometry in single point scanning arrangement, which enables lateral critical dimension measurement with greater accuracy, but still diffraction limited to the spot size of around 1 micrometre.

[24] A. Rodriguez, M.C. Morant.Minana, A. Dias-Ponte, A. Martinez-Calderon, M. Gomez-Aranzadi, S.M. Olaizola, Femtosecond laser-induced periodic surface nanostructuring of sputtered platinum thin films, Applied Surface Science 351 (2015) 135-139

[25] A. Martinez-Calderon, A. Rodriguez, A. Dias-Ponte, M.C. Morant.Minana, M. Gomez-Aranzadi, S.M. Olaizola, Femtosecond laser fabrication of highly hydrophobic tainless steel surface with hierarchical structures fabricated by combining ordred microstructures and LIPSS, Applied Surface Science 374 (2016) 81-89

[26] E. Bugnicourt, T. Kehoe , M. Latorre, C. Serrano, S. Philippe and M. Schmid, *Recent Prospects in the Inline Monitoring of Nanocomposites and Nanocoatings by Optical Technologies*, Nanomaterials 2016, Review, 6(8), 150; doi:10.3390/nano6080150

[27] THIME, Thinfilm measurements on organic photovoltaic (OPV) layers; FP7Grant Agreement no: 315665; 2012-2014

[28] dr. schwab GmbH, IQPC – Integrated Quality and Process Control. http://www.schwabinspection.com/index.php?puid=1&pageid=39

[29] Nanometrics Impulse, http://www.nanometrics.com/products/IMPULSE.html

Scatterometry: It is possible to obtain higher resolution in lateral dimension measurements using optical techniques based on scatterometery and diffractometry, which can be applied to periodic structures of the type being developed in OLEDSOLAR. Scatterometry uses the instrumentation of a spectroscopic ellipsometer, along with complex mathematical modelling[30] and is widely used in the microelectronics industry, for example the Nanometrics Atlas[31], and the Nova T600[32]. However, the complex instrumentation and high-power data processing required mean that the technique has not been applied inline. Coherent Fourier Scatterometry, has been developed to exploit the smaller spot size that is possible using laser light, but is currently not deployed inline[33].

***Diffractometry*:** Diffractometry using multiple diffraction orders imaged in the far-field is a versatile technique that has potential to provide comprehensive information regarding the 3 dimensional features of textured samples. Furthermore, when combined with different light source, it covers a broad resolution range: UV light can detect structures with period in the few hundreds nm range, whereas cost competitive blue laser can be used for structures with period between 500 and 700 nm. It has been effectively implemented in several versions for characterising the quality of the grooves in DVD and Blu-Ray discs, for example dr. schwab DOMS[34], and Singulus Mastering B.V.[35] However the optical system is quite complex and so it hasn't been employed in an inline configuration.

Confocal Microscopy: may be used to obtain spatial resolution below the diffraction limit with resolution of 0.09um reported[36], however once again it is not implemented inline.

Another salient feature of this technique is that it provides an image of the surface structure, from which specific, local dimensions can be calculated, whereas for the current technology under development, it is desirable to obtain a general overview of the shape and common dimensions repeated in the periodic pattern, especially where the sizes of individual lines can very over a range. For this reason, it is more effective to employ a metrology that makes use of the structure periodicity, such as diffractometry or scatterometery, and provides a description of key features, including height, width, sidewall slope and symmetry. Diffractometry is the more attractive option of these two for inline process monitoring since it can provide simple information quickly with less data processing.

Inline inspection of a huge variety sample parameter of interest ends their spatial distribution. Common technologies for the evaluation of barrier performance are extremely time consuming. Reliable measurements of the WVTR of ultra-barrier webs using diode laser spectroscopy (HiBarSens®) need days or some weeks (10^{-6} g m^{-2} d^{-1}). Totally new concepts based on HSI are just demonstrated for a fast prediction of the performance of barrier webs (in the range of 10^{-2} g m^{-2} d^{-1}) with sufficient accuracy.[37]

Progress beyond the State of the art:

The challenge to progress beyond the state of the art will be to effectively implement diffractometry for inline process monitoring. This will require the design of compact, high numerical aperture optics to capture the maximum number of orders possible. Similar optical systems have been used to apply sub-wavelength optical diffraction metrology[38],[39] to charactrise structures as small as 50nm in periodic patterns.

As such, IRIS will start by calibrating the system on parts of known geometries with several patterns required for obtaining the effects for each application. Variations of the basic system may be developed, having adjusted light sources and optics to the resolution and spot size needed, and for compatibility with flat surfaces, used in OLED and OPV industries. These will be integrated in the related production lines to provide dimensional information on the structures in real time. Resulting data can be used to feed the processing models where appropriate.

The transfer of the lab-scale approach of HSI based barrier evaluation to a continuous inline evaluation of barrier webs is highly ambition. Beside the hardware implementation (camera, lighting) the development of reliable robust prediction models (based on combination of deep learning algorithms and imaging processing) are extremely challenging. It should also be noted that the approach has to be developed and tested for ultra-barrier webs (10^{-4} g m^{-2} d^{-1} and better) with corresponding ground truth measurements. The inline HSI implementation will be accompanied with large area calcium mirror tests and large area OLED coatings on to be evaluated barrier films in roll-to-roll fabrication. The direct correlation of the device degradation behaviour and OLED lifetime performance from classical storage climate chamber will be applied as an input for the inline HSI algorithm. Yield and device reliability relevant

[30] http://www.dfm.dtu.dk/index.php?id=309&L=-1%27
[31] Nanometrics Atlas, http://www.nanometrics.com/products/atlas-II.html
[32] Nova T600, http://www.novameasuring.com/t600.html
[33] S.Roy, A.C. Assafrao, S.F. Pereira, H.P. Urbach, Coherent Fourier Scatterometry for detection of nanometer-sized particles on a planar substrate surface, Optics Express 22 (2014) 13251
[34] Dr. schwab DOMS XE-blu, http://www.schwabinspection.com/index/dr.schwab+Inspection+Technology/DOMS+Optical+Disc+V2.html
[35] http://www.singulus.com/en/innovative-technology/surface-engineering/crystalline.html
[36] Sensofar S Line, http://www.sensofar.com/metrology/sneox/
[37] P. Wollmann, F. Gruber, W. Grählert; „Anordnung zur Bestimmung der Permeationsrate einer Probe", 2015, Patent, erteilt; DE 10 2015 213 974; CT/EP2016/067019
[38] *Kehoe, Microelectronic Engineering, 86, 1036 (2009)*
[39] *Silver, Appl. Opt., 46 (20), 4248 (2015). - Kreuzer, Proc. SPIE 9424, 942426 (2015); Kreuzer, Proc. SPIE 9628, 96281Q (2015)-957 (2005)*

issues caused by the barrier films substrate and top encapsulation will be carefully validated and in close relation with the HSI inspection system.

Progress beyond state of art in automation and control:
The challenge to progress beyond the state of the art will be to effectively implement smart sensors for inline process monitoring and control them in closed loop.

These will be integrated in the related production lines to provide critical parameters information on the components in real time. Resulting data can be used to feed the processing models where appropriate.

Thanks to the application of the novel inline monitoring system, OLEDSOLAR project will reduce the current processing cost and time to achieve a cost-effective process, up scalable
to mass production.

State of the art and Progress beyond state of the art: (for recycling and re-use)

Optimisation and understanding of advanced products such as OLED and organic solar cell recycling requires deep systemic knowledge and understanding of the physics-based link between product design, material applications, potential reuse of components and combinations with the final treatment process technologies in the entire recycling system or chain. Rapidly changing product design and product replacement poses challenges to physical recyclers to have recycling technology keeping pace with rapidly changing product properties and material usage, which poses different demands on sorting technology development and plant configuration in order to most optimally recover materials[40].Today, limited information is available on the internal reuse and recycling of components along the manufacturing process of commercial photovoltaic panels already installed at European scale. However, all photovoltaic modules available on the European market can be disposed of, notwithstanding the type of technology used. Most parts of a solar module can be recycled, including glass, semiconductor materials, ferrous and non-ferrous metals. The modules present on today's market belong to two different categories, silicon and non-silicon based, which determine the recycling process to be used[41].

- For silicon-based modules, aluminium frames and junction boxes are dismantled manually at the beginning of the process. The module is subsequently crushed, and its several components are separated, allowing recovering up to 80% of the panel.
- Non-silicon-based panels require the use of diverse recycling technologies. Cadmium telluride (CdTe) panels e.g. – a particularly common type – are first crushed into different fractions, much like non-silicon modules. But they also use chemical baths to separate the various semiconductor materials, allowing for the recovery of 95% such components. Recycling technologies for this type of panels have been widely increasing in recent years. For copper indium selenide (CIS) and Copper indium gallium (di)selenide (CIGS) photovoltaic modules similar chemical bath treatments apply.

Even different EU funded projects have been focused in the development of new OLED and flexible solar products at pilot and semi-industrial scale (such as FLEXOLIGHTING, IMOLA or PI-SCALE) by using techniques such as vacuum thermal evaporation and shadow masking, organic vapour phase deposition (OVPD), transfer- printing and/or inkjet printing, the recyclability and reusability potential of novel OLED and flexible solar products, materials and components will be tackled in OLEDSOLAR project by using an innovative approach aimed at ***(1)*** defining the optimal reusability route for valuable wastes along the manufacturing process as well as ***(2)*** the evaluation and selection of the most feasible recycling solution according to the industrial needs.

Innovation Potential: As an alternative to the current attempts to identify opto-electronics devices faults through largely inefficient empirical and visual methods, OLEDSOLAR is based on the idea of multi-system reliability and control theory: model-based process approach; however, the project clearly advances the state of the art by introducing a series of new models, algorithms and technologies enabling the application to OLEDs and solar cells development processes with a degree of complexity never handled before. This return to 'basics' or 'first principles' approach, on regard to accurate measurement and control, coupled with the utmost knowledge and expertise of the consortium on latest manufacturing, inspection, machine learning and automation technologies will enable OLEDSOLAR to give a truly innovative response to a long, well established problem of our industry.In fact, OLEDSOLAR will introduce radically new methodologies in the design and set up of control systems for

[40] E. Worrell and M. A. Reuter (2014) Handbook of recycling. State-of-the-art for practitioners, analysts and scientists. ISBN: 978-0-12-396459-5.
[41] http://www.solarwaste.eu/collection-and-recycling/

manufacturing processes. If until now, control systems are specifically developed for each machine, by ad hoc trained personnel, OLEDSOLAR will leverage the expertise of multidisciplinary consortium from the start of the development. This will enable the introduction of new control concepts to target variables critical for production, concepts which will be efficiently implemented through advanced techniques of automation and control, including special optical measurements.Ultimately, this well enhance technological knowledge enabling a new framework of relationships between customers and manufacturers, since system will be able to accurately program and schedule the manufacturing operations to respond to the requirements, whereas the first will be able to rely that customer's expectations are met in good quality and time. This benefit will be invaluable within the framework of markets of ever growing competitiveness (for instance, for automotive, healthcare, aerospace), in which normally, the handling of inefficiencies during set up and production represent the border between failure and success.Hence, the successful demonstration of the architecture in this multi-physics scenario will enable the OLEDSOLAR architecture to be applied to other processes in the future aiding its penetration in our manufacturing market even further.

Relevant Projects Comparison:

Table 4: Relevant projects comparison with OLEDSOLAR approaches

Relevant Projects	Differences vs. OLEDSOLAR
PI-SCALE (H2020)-*Bringing flexible organic electronics to pilot innovation scale*-(PI-SCALE) is dedicated to integrating existing European infrastructures into a "European flexible OLED pilot line", which will operate in an open access mode and serve customers from along the value chain with individual product designs, validation of upscaling concepts, and system-level flexible OLED integration.	OLED specific activities in PI-SCALE project and limited from inspection and metrology point of view, where OLEDSOLAR solution will provide optimised processes, metrology and automation for OLEDs (both rigid and flexible) within R2R and S2S setup as well as for solar cells manufacturing. The development activities from PI-Scale project will be transferred and fed into OLEDSOLAR proposed activities.
CLEAN4YIELD (FP7)-*Contamination and defect control for increased yield for large scale R2R production of OPV and OLED*-The overall objective of Clean4Yield has been the development and demonstration of technologies and tools for nano-scale detection, cleaning, prevention and repair of defects and contaminations in nano-scale layers within R2R production processes for OLED and OPV.	Inspection specific project activities with surface defects detection, where OLEDSOLAR proposed inspection system will cover the manufacturing chain considering interlayer inspection, correlations of measurements and device level NDT. A hybrid system will allow to monitor and inspect many parameters allowing end user to reduce waste and increase product yield.
Flex-o-Fab (FP7)-*Pilot-scale hybrid roll to roll/sheet to sheet manufacturing chain for flexible OLEDs*-The overall objective of the Flex-o-Fab project is the demonstration of a reliable manufacturing process for OLED lighting foils based on the manufacturing of glass-based bottom emissive small molecule OLEDs and use these as a basis to develop a pilot series manufacturing process for flexible OLEDs.	OLEDSOLAR is dedicated to OLED displays and solar cells high yield production. The processes proposed will also allow process reconfiguration time reduction as well as significant recycling. The project will be built upon existing R2R and S2S process steps at TRL4 and will develop TRL6 systems for final scale up and market transfer.
LOMID (H2020) - *Large cost-effective OLED micro displays and their applications*-The project will define pathways for the manufacturing of flexible OLED micro-displays with large area (16 mm x 20 mm, screen diagonal of 25.4 mm) at yields (>65%). This will be achieved by developing a robust silicon-based chip design allowing high pixel counts (1024x1280 (SXGA)) and high spatial resolution (pixel sizes of 10 µm x 10 µm corresponding to 2000 ppi).	Th proposed technologies in OLEDSOLAR will allow manufacturing of large display OLEDs with improved barrier performance, reduced permeation and high quality and devices lifetime. The proposed metrology systems proposed in project will allow high level of product reliability and quality as well.
FLEXOLIGHTING (H2020)- The flexo-lighting programme is focussed on research and innovations on materials, processes and device technology for OLED lighting with the intention of building a supply chain within Europe. The aim is to realise OLED devices over a large area/surface with high brightness, high uniformity and long-life time. A demonstrator will be built and delivered at the end of the project.	OLEDSOLAR project is focused on production and manufacturing processes and metrology systems that will ultimately allow development of long-life OLED devices and efficient solar cells. The most critical steps in production and metrology has been targeted in the project with an impact on yield, process reconfiguration time and recyclability.

Fast2Light (FP7)- *High-throughput, large area and cost-effective OLED production technologies*-The overall objective of Fast2Light is to develop novel, cost-effective, high-throughput, roll-to-roll, large area deposition processes for fabricating light-emitting polymer-OLED foils for intelligent lighting applications. The scope of the project comprises all the layers that are part of an OLED lighting foil.	OLEDSOLAR project offers a novel, cost-effective, high-throughput, roll-to-roll as well as S2S, large area deposition processes for barrier films for OLEDs, beside doing performance analysis and qualification of barrier performance. The scope of the project involves metrology, inspection and quality assurance system for OLEDs fabrication process as well.

Patent search:

The patent search has been carried out showing some of the relevant patents below:

- **US20050172897A1**-A method and apparatus is described using atomic layer deposition to form a barrier layer onto a substrate. The coated substrate exhibits reduced permeation to oxygen and water vapor.
- **US20110033957A1**-nvention generally relate to systems, apparatuses, and methods used to form solar cell devices using processing modules adapted to perform one or more processes in the formation of the solar cell devices.
- **US20080264477A1-**Methods for manufacturing three-dimensional thin-film solar cells
- **US20070227586A1-**Detection and ablation of localized shunting defects in photovoltaics
- **US20100197051A1-**Metrology and inspection suite for a solar production line
- **US20120103383A1-**Photovoltaic Device and Method and System for Making Photovoltaic Device
- **US6063995A-**Recycling silicon photovoltaic modules
- **US20050253142A1-**Solar cell and its manufacturing method
- **US20080251120A1-**Thin Film Solar Cell and Manufacturing Method
- **US5241482A-**Monitoring system for automated assemblies

The analysis shows OLEDSOLAR proposed solutions are not in conflict allowing the freedm to operate, and also innovative nature is proved considering not many integrated solutions are developed to achieve targeted performance set in OLEDSOLAR.

2 Impact

2.1 Expected impacts

Project contribution to the expected impacts:

IMPACT 1: 15% yield improvement because of the introduction of new sensor equipment, related metrology and automatic non-destructive control:

The proposed processes in OLEDSOLAR including (1) optimised, enhanced and scaled up manufacturing processes for OLEDs, OPVs and CIGs targeted at high yield, low cost production, (2) Development of methodology for the automatization of the opto-electrical characterization of solar cells and OLEDs, (3) Optimisation of inspection methods (hardware and software) for in-line monitoring, inspection and quality control in R2R environment for (CIGS and organic) solar cells & OLED displays,(4)Development of an hybrid measurement optical module (Photoluminescence, optical diffraction, Raman spectroscopy, Spectroscopic ellipsometry) and (5) automated manufacturing process and enhanced machine learning defect analysis being developed in project will not only increase the yield of production by efficient production process but also will increase product lifetime, efficiency and performance of opto-electronic devices.

In summary Production yield improvement factors and relevant OLEDSOLAR proposed solutions are summarised below: **Table 5: Impact on production yield and OLEDSOLAR relevance**

Production Yield Criteria of OLEDs, OPVs and CIGs (Process yield × Product yield)	OLEDSOLAR proposed solution and Impact on Yield (>15%)
Process Yield • Deposition quality (thickness, composition, uniformity) • Encapsulation quality • Thickness and linewidth • Manufacturing quality and uniformity of layers • Glass Cleanliness • Broken/Cracked panels	• Advance high speed, cost effective SALD and PESALD deposition processes resulting in efficient barrier coatings and encapsulation. • Optical inspection for thickness measurement, manufacturing quality analysis and layers uniformity check, • NDT for cracks and mechanical defects testing • Photo-Luminescence testing for optical performance testing

• Shorted Panels • Planarization quality • Optical performance Product Yield: • Mechanical defects • Back-End module assembly • Electrical connections quality • Efficacy • Lifetime	• Functional testing and printing quality testing and measurements, • AI driven machine learning based defect analysis and processing software, • Sensors system for automation and control of manufacturing sequences. • Sensor based modules performance testing Automatic registration systems will reduce the number of misaligned components in the start-up phase and thus reduce the time when changing the component type.

IMPACT 2: 15%-time reduction for reconfiguration of key process tools in a production line due to change of type of component: OLEDSOLAR proposed solution will involve intelligent decision management software enabled by AI that will fed devices requirements into production process in fast and effective way to allow adjustment of inspection and manufacturing steps sequence. Further, automated material handling and manufacturing equipment being optimised in OLEDSOLAR e.g. Click and Coat system being developed by Coatema for opto-electronic devices manufacturing has the potential to allow process reconfiguration time reduction by 40%. This will allow industrial partners to customise their products and run different components batches in effective way without production downtime.

Table 6: Impact relevance to OLEDSOLAR project

Process Reconfiguration Requirements based on devices changing specification	OLEDSOLAR proposed solution Impact (15% reduction potential in process reconfiguration)
• Change of layers thickness • Requirements of customisation • Adaptability of process • Demand based processing tooling • Intelligent resource efficient production route • High speed processes and inspection • Low defect rates • Low tooling maintenance • High uptime of equipment • Less processing steps	OLEDSOLAR proposed solution involve intelligent decision management software capable of planning production plan for set of batches of opto-electronic components. Based on requirements tooling and setups can be devised and reduced depending upon need and process can be reconfigured quickly. Tooling for deposition, automation in production line and sensor enabled monitoring will allow faster and productive manufacturing. In-line inspection will reduce offline quality testing, post treatment and pre-treatment setups. Automated Click and Coat system for OLEDs and solar cells manufacturing

IMPACT 3: A tangible part (> 10%) of the production cost of the parts should originate from recycled products and materials:

Development of green-solvent-processable OPVs in terms of design considerations, material application and solvent selection; Fabrication of efficient organic solar cell utilizing scalable deposition methods using halogen-free solvents. OLEDSOLAR project propose and compare recycling techniques for opto-electronic parts and will devise repair and re-use strategies allowing recycling of more than 98% of discarded devices and components resulting in more than 10% of production cost covered through recycling into the production line. Material challenges will be addressed, and halogen free options will be considered to reduce waste material.

ASPF, SOLIBRO AND INURU has strategic objective of reducing production waste and developing resource efficient product development strategy.

Impact on Project End Users:

The topic relevant impact on project end users and on their devices/ production lines is detailed below:

Table 7: Impact of OLEDSOLAR on End users

End Users	Relevant Impact on their production and devices manufacturing through OLEDSOLAR
ASPF (for OPVs)	**IMPACT 1**-Current production yield of ASPF for OPVs is 85-90% which result in 10% total production loss due to quality issues, absence of inline metrology and manual manufacturing processes. By implementing patterning, coating, automation of processes and encapsulation techniques developed in OLEDSOLAR, it will result in yield improvements of more than 15% for OPVs production of ASPF achieving their targeted yield of >98%.
	IMPACT 2-Laser structuring allow ASPF customisation of devices but other process e.g. manual screen printing and semi-automated production line result in significant process reconfiguration. The desired automated manufacturing process and AI enabled decision

	management software being developed in OLEDSOLAR will allow process adjustments as per devices need thus reducing process reconfiguration time by more than 15%.
	IMPACT 3- ASPF losses 5-10% of total production cost due to faulty parts and devices manufacturing and requires printed silver in OPVs to be recycled. OLEDSOLAR proposed processes will allow to recycle and re-use more than 98% of parts into the production line resulting in production cost being covered through recycled parts and materials by 10%.
Solibro (for CIGs)	**IMPACT 1-** Solibro is working into developing new materials-based CIGs as well beside traditional material layers, this requires inline metrology to achieve high yield and production efficiency of manufactured solar cells. OLEDSOLAR proposed photo-luminescence, spectroscopic techniques and NDT techniques will help Solibro to achieve high yields improvement (>15%)
	IMPACT 2- Automated system and intelligent software will help Solibro' s current production setup to adjust processes as per devices need and demand resulting in more than 15% reduction of process reconfiguration.
	IMPACT 3- Solibro' s target is to achieve 100% recycling or repair process in their production line, the developed processes in OLEDSOLAR will support CIGs defected devices recycling and generation of more than 10% of production cost through recycling and re-use saving resulting in production efficiency and operating cost.
DTF (for thin films)	**IMPACT 1-** Current production yield is 50% for their films, due to failure in meeting specification resulting production losses and complaints. By implementing inline metrology system as proposed in OLEDSOLAR this issue will increase yield by more than 15% for film manufacturing to achieve their targeted production yield of >65%.
	IMPACT 2- Average time for process reconfiguration for DTF is 1 day, the configuration can be done quickly depending upon quality aspects of films if monitored accurately, thus there is a potential of reducing it by more than 15% through inline metrology, automation, barrier film qualification and defect recognition software proposed in OLEDSOLAR.
	IMPACT 3- Due to waste DTF loss 4500 euros/10.000m2 on films, as currently, the film quality is compromised frequently due to lack of quality inspection system inline capable to analyse faults, OLEDSOLAR proposed inline metrology will enable DTF to achieve their targeted more than 90% recycling of films resulting in creating more than 10% of production cost through recycling and re-use.
INURU (for OLEDs)	**IMPACT 1-**Current production yield is 90% with a need to achieve more than 95% production yield. OLEDSOLAR proposed solution will allow the to enhance their yield through improved inkjet process, improved barrier foils and inline metrology resulting in achieving their targeted production yield and reduction of cost per device.
	IMPACT 2- Due to various application and type of OLED devices INURU is manufacturing in their production line, significant time is spent in process reconfiguration as per devices need and specification. The automated tool kit including intra equipment automation and Ai based decision management software reading devices requirement will select and run process automatically resulting in process adaption and reduction of process reconfiguration time by 15%.
	IMPACT 3- Due to high value devices, INURU wants to recycle and re-use 100% of faulty parts through improved recycling techniques and repair procedures. OLEDSOLAR proposed solution of recycling will enable OLED devices to achieve their target and generate more than 10% of production cost through recycling.

Project outcome and contributing factors to achieve targeted impact figures:

Impact factor on scale of 1-5 is considered for every outcome:

Table 8: Impact Matrix with expected impact and project objectives

OLEDSOLAR Project Outcome	Impact 1	Impact 2	Impact 3
Adaptive material handling system (Objective 1)	2	5	
Optimised printing process for solar cells (Objective 2)	4		
Deposition processes (Objective 3,4,5)	5		
Automated lamination and coating processes,(Objective 6)	3	5	

Optimised manufacturing process and validation of barrier films (Objective 7,11)		4		
Optimised inline hybrid inspection system (Objective 8,10)		5	3	3
Optimised in-line NDT testing (Objective 9)		3	2	3
Functional testing and measurements (Objective 12,13)		3	2	2
Machine learning based inspection data processing and automated defect detection software(Objective 14)		5	4	
Sensor-based network for automation and monitoring (Objective 15)		4	5	
Intelligent AI based decision management and control software (Objective 16)		4	5	3
Scalable recycling solution including pre-treatment strategy, sorting and re-use strategies(Objective 17)		2		5
Repair and re-use strategies (Objective 18)		2		5
Total Impact (Cumulatively the impact figures will be higher because of interlinked objectives and impacts)	Minimum Improvements	>15% Yield improvement	>15% process reconfiguration time reduction	>10% the production cost of the parts from recycled products and materials
	Cumulative improvements	>40% Yield improvement	>25% process reconfiguration time reduction	>20% the production cost of the parts from recycled products and materials

Enhancing Innovation Capacity:

The main value proposition of the OLEDSOLAR is based on the possibility to manufacture personalized and customized, flexible, more functional materials opto-electrical products e.g. OLEDs, CIGs and OPVs for different application using efficient, adaptive and cost-effective manufacturing processes and quality testing techniques. Improved manufacturing processes and inline of the manufacturing process will improve the yield and reduce the final cost of the products. The stake is major for Europe: relocate some tech industries by decreasing the production cost of high-performance components. The OLEDSOLAR project will be focused on innovating critical manufacturing processes along with advanced inline quality control and NDT systems development for production of efficient optoelectronics parts, which will be a major opportunity for European SMEs and industries to acquire high yield and less costly production processes. Allowing the possibility for companies of all sizes to develop and fabricate high-performance opto-electronics products at low cost with a high reproducibility and reliability will lead to many innovations in Europe.

Figure 49 Project scope and application

The OLEDSOLAR technology will allow production of individual or small series opto-electronics parts with a simplified and shorter supply chain. This would be a major benefit for SMEs to slot into a value chain currently locked by big companies. By reducing the

number of operations, simplifying manufacturing process and its control, and reducing the number of appliances required to manufacture and inspect opto-electronic product with the right features, we will be able to spread the technology at table-top level for the European companies.

The methodologies developed in the OLEDSOLAR project in terms of manufacturing, quality assurance & control will strongly improve the quality, the reliability and the cost of the products which is essential for a successful implementation of the OLEDSOLAR technologies.

Creation of market opportunities, strengthening of competitiveness and growth of companies:

OLEDSOLAR solution will enable high quality manufacturing of OLED devices that have numerous existing and emerging applications as depicted in figure . The outcome of this project will enable further, adaption and feasible cost effective production of rigid and flexible OLEDs for various application:

Some of the market application of OLEDs with market share is depicted below.

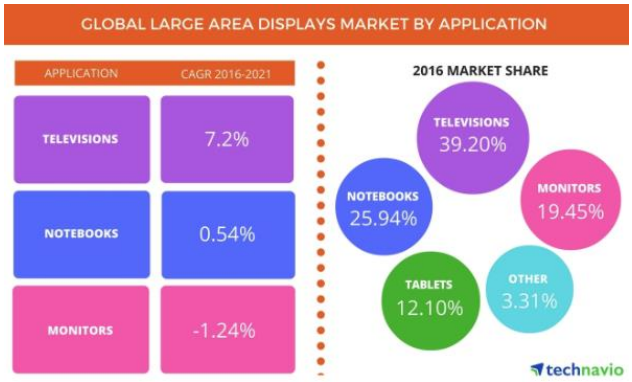
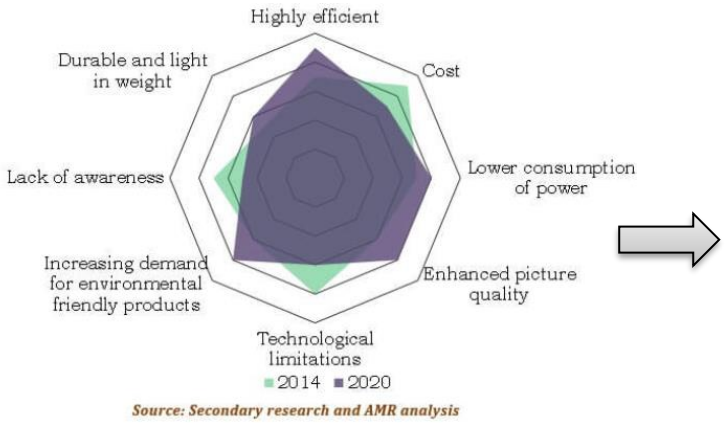


Figure 50 OLED applications

OLED market is expected to be valued at 40 Billion euros by 2023, growing at a CAGR of 15.2% between 2017 and 2023[42]. Commercialisation and exploitation of proposed production systems in OLEDSOLAR will allow involved partners to capture market share in potentially emerging market. Considering the fact that OLEDSOLAR is aligned to address market success factors for successful commercialisation as depicted in figure below:



OLEDSOLAR addressing those factors:

- OLEDSOLAR proposed activities will help to reduce cost of production resulting in highly efficient, durable and light weight devices.
- Dedicated WP7 activities will disseminate the project thus raising awareness of developed technology and adoption,
- Recycling and re-use activities proposed will result in environmental friendly product development,

Figure 51 OLEDs market success factors and how OLEDSOALR addressing those factors

INURU being end user and manufacturer of OLEDs will be able to improve its processes and develop better OLED product enabling them to enhance their sales through reduction in cost of manufactured devices and extension in lifetime and efficiency.

Solar Cell Technology: Further, in solar cells technology including thin film and organic solar cells, the demand of such application is increasing due to need of creation of eletricity through renewable sources. The figure below shows the demand and supply situation and emerging efficiencies of potential solar cell technologies .e.g. CIGs.

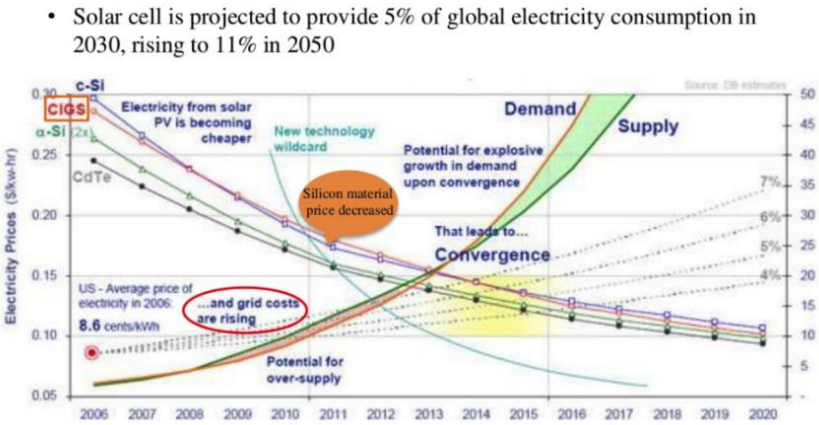


Figure 52 Solar cells potential (Source: IEA solar PV roadmap)

The development of inline metrology and optimized manufacturing processes of OLEDSOLAR will enable to achieve higher efficiencies in solar cell, Solibro (End user in OLEDSOLAR consortium) has broken efficiency records – double glass CIGS solar modules at 18.72 percent and sputtering CIGS solar modules on flexible substrate at 17.88 percent . Their issues related to metrology, absorber layers and induction of new material in their cell

[42] https://www.marketsandmarkets.com/PressReleases/global-OLED-market.asp

technology requires inline advanced metrology systems as proposed in OLEDSOLAR to achieve even higher efficiencies in solar cells. Further, CIGS market share is expected to exceed 6 billion euros by 2024[43]. Which presents an opportunity for exploitation of OLEDSOLAR solution to emerging industry where, EU can create a competitive position.

Moreover, O**rganic Photovoltaics (OPV) market is** expected to reach €**350 million by 2021**[44]. The specific printing processes, automation , inspection and quality control and packaging techniques proposed in OLEDSOLAR will enable high efficiency production of OPVs and will enable ASPF (end user in project and OPVs manufacturer) to improve their devices by product diversification , production cost reduction and customisation.

One of the most emerging application of solar cells is in the Building-integrated photovoltaics (BIPV). According to market report BIPV technologies will grow from $2.4 billion in 2016 to $4.3 billion by 2021 with a compound annual growth rate (CAGR) of 12.2% for the period of 2016-2021[45].

This presents a potential opportunity allowing solar cell manufacturing partners in OLEDSOLAR (ASPF, Solibro) and V industry in EU to diversify products and exploit benefits of OLEDSOLAR solution enabling high yield, high efficiency and low cost manufcaturing of solar cells. The proposed solution will also enable commercialisation partner(SMIT TS,IRIS,COATEMA,ADVENT) to capture potential OPVs production and inspection systems market share.

The market opportunity shows significant potential of OLEDSOLAR solution in manufacturing high value products and devices in cost effective and productive way resulting in increasing capacity, innovation potential and economic benefit for companies and partners involved in the value chain.

Financial projection of sales of OLEDSOLAR systems:

OLEDSOLAR systems will be further, scaled up post project and as forecasted below in 5 years of commercialisation will generate economical returns for the consortium. OLEDSOLAR commercialisation will generate 117 million euros in revenues, with 52 million euros in gross profits within 5 years considering profit margin of 45%. Return of investment (ROI) on EC investment will be 5:1.

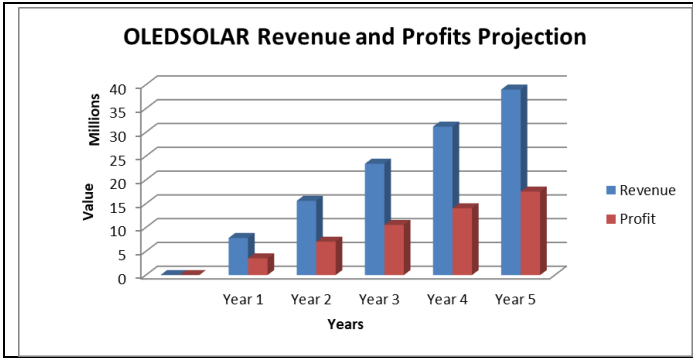
Table 9: Financial projection of OLEDSOLAR

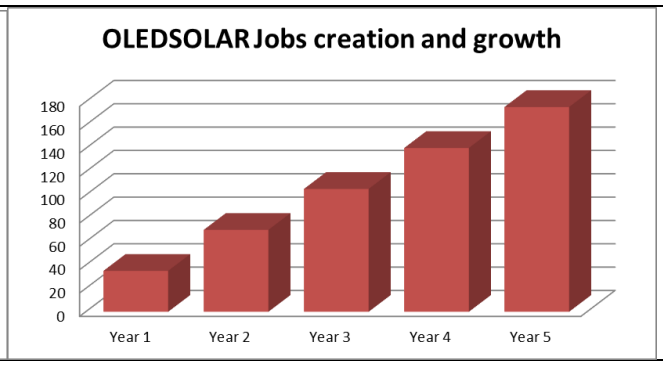
Years	2026	2027	2028	2029	2030	Total
	Year 1	Year 2	Year 3	Year 4	Year 5	
OLEDSOLAR Systems Price (€) approx.	1,560,000	1,560,000	1,560,000	1,560,000	1,560,000	
Deposition Equipment (€)	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	
Inline hybrid Inspection System (€)	150,000	150,000	150,000	150,000	150,000	
Inline NDT systems (€)	100,000	100,000	100,000	100,000	100,000	
Automation hardware and sensor kit (€)	50,000	50,000	50,000	50,000	50,000	
Software license (€)	10,000	10,000	10,000	10,000	10,000	
Printing and coating equipment (€)	250,000	250,000	250,000	250,000	250,000	
Targeted Market Share	0.50%	1%	1.50%	2.00%	2.50%	
Targeted number of sales	5	10	15	20	25	75
Sales (million €)	7	15	23	31	39	117
Profit Margin	45%	45%	45%	45%	45%	
Gross Profit (million €)	3	7	10	14	17	52
Jobs created in consortium (/100k profit)	35	70	105	140	176	527
Forecast of Revenue, Profit and Jobs creation and growth post commercialisation of OLEDSOLAR						

[43] https://www.gminsights.com/industry-analysis/thin-film-solar-cells-market

[44] http://www.marketresearchstore.com/report/global-organic-photovoltaics-opv-market-rport-2017-154896

[45]https://www.bccresearch.com/market-research/energy-and-resources/building-integrated-photovoltaics-markets-report-egy072C.html





Social Impact (Healthier work environment): OLEDSOLAR solution will enable less potential safety risks in the factory floor due to a reduced need for human interventions. Human interventions are risky, as injuries may occur when a machine stops. Less human involvement in the shop floor, means less exposure to a harmful environment. Less creation and handling of waste also contributes to a safer and healthier workplace.

EU Competitiveness and SMEs:

Taking into consideration the global aspects and specificity of the companies. **OLEDSOLAR project** aims to Strengthen the Competitiveness and Growth *of Companies* through developed Innovative developments also addressing and meeting the needs of EU and global consumer Markets providing a competitive advantage to Europe. The impact on the competitiveness of all the manufacturing industry will be important since there are other sectors with similar circumstances and using similar manufacturing processes and inspection systems. So, at the end the impact will spread to the manufacturers of opto-electronic products.

Barriers/Obstacles and framework conditions:

Table 10: Barriers to effect impact achievement

Barriers/Obstacles, framework conditions	Effect on achievement of the OLEDSOLAR Impacts	OLEDSOLAR measures to address Negative Impact factors
High costs of innovation activity, the lack of an appropriate source of finance and perceived excessive commercial risks are recognized barriers for every solution.	OLEDSOLAR project market penetration will get effected in short term	Overcoming these limits by developing a more effective and efficient technological solution both in terms of quality, resource efficiency and energy consumption, with a high coefficient of market exportability and replicability. OLEDSOLAR solution can be characterised/ adapted according to industrial/market need and the price customers are willing to pay.
OLEDSOLAR system is highly innovative and diverse and often production methods are based upon experience rather than technology.	Initial inertia to adapt the system in production line.	Low cost, high speed and high reliability manufacturing and inspection system will attract end users to adapt the OLEDSOLAR system in production operations.
Certification, standards and safety regulations on factory floors and production facilities	Can be expensive and time consuming and decrease the extent of industrial exploitation.	OLEDSOLAR proposed system from design to scale up phase is being developed considering standards, regulations and any certification that maybe required. OLEDSOLAR project will feed standards of factory and workers safety as well as opto-electrical devices process architecture.

2.2 Measures to maximise impact

a) Dissemination and exploitation of results

Dissemination, Communication and Exploitation activities(will be reported in PEDR,activities in WP7) are of paramount importance in OLEDSOLAR project, in order to maximize its impact and trigger effects across the project's entire range of target audiences. In particular, the OLEDSOLAR consortium, by fully recognizing the above, will implement a dedicated dissemination and exploitation strategy, predominantly aiming at ensuring:

- The effective and sustainable dissemination of the OLEDSOLAR project generated knowledge and technologies within the entire community, and through the implementation of *suitable* and *specialized* dissemination and communication activities for each of the project's identified target groups and end users.
- The exploitation of the project's results by the European Manufacturing Industry so as to maintain and reinforce technological advantage over the competition from outside Europe.
- The interconnection with other industrial sectors with the potential to exploit the findings and outcomes of the project.
- The conveyance of new knowledge into the engineering education base provided by the University partners to meet the evolving skill needs of the sector.
- The possible exploitation of the OLEDSOLAR project results and foregrounds in other sectors, e.g. nano-electronics and nano-scale components development.

Dissemination is a significant tool that will be used to connect people (i.e. consortium members, general public, specific target groups of stakeholders in the related industrial/scientific fields) to the achievements and activities performed within this project. **Exploitation** activities involve all actions related to the use of the knowledge generated within the project. Consequently, **by effectively and strategically disseminating and exploiting the project's results, greater public awareness is created as well as knowledge sharing, transparency and education are promoted.** According to Eurobarometer (December 2013 issue), close to two-thirds (64%) of Europeans believe that it is important *"to increase the support for research and development policies and turn inventions into products".* Considering that EU-funded activities and projects distinctly contribute to the creation of new jobs, novel technologies and improve the citizens' quality of life, public interest for their findings and achievements is increasingly growing over time. Moreover, since the largest and foremost financer of such projects is the European taxpayer, it is an imperative precondition to ensure: *i) maximum return on the investment*, and *ii) full openness about the actions financed.*

In this context, a tailored dedicated plan for the dissemination and exploitation of the OLEDSOLAR project results has been prepared and is detailed below. This draft plan summarises the consortium's strategy and concrete foreseen actions (to be performed during and beyond the end of the project) to disseminate, exploit and protect the new knowledge produced within OLEDSOLAR.

Plan for the dissemination and exploitation of the project's results (PEDR) : The primary objective of the draft OLEDSOLAR Dissemination and Exploitation Plan is to prudently and timely identify and organize the activities to be performed (during and after the project), in order to maximize its influence and while taking into account the dissemination needs of the project at each stage of its lifecycle, as well as the specific technical, market, organisational issues and interests of each of the various pre-defined OLEDSOLAR target groups/end users. Consequently, the **main aims** of the planned dissemination and exploitation approach can be summarized as follows:

- **Inform** about the OLEDSOLAR activities to stimulate the participation of SMEs, Academia, Industry, Research establishments, civil society and their networks. Organization *of (and participation to) events, workshops and seminars;*
- **Raise Awareness** -Present the project, its main objectives and expected impact (e.g. OLEDSOLAR public website, project leaflet, poster and newsletter, etc.)
- **Networking** -Exchange experience with other projects related to OLEDSOLAR so as to join efforts, minimize duplication and maximize exploitation potential
- **Disseminate Knowledge** -Regularly provide information about the OLEDSOLAR outputs through several channels (e.g. Horizon the EU Research and Innovation Magazine, research*eu results magazine, research*eu focus, etc.)
- **Support OLEDSOLAR Exploitation** - Pave the way for a successful exploitation of the project results by addressing the full range of potential users and uses, including research, commercial, investment, social, environmental, policy making, setting standards, skills and educational training.

The following table provides a list of the main dissemination activities in OLEDSOLAR. More details on communication activities can be found on section b.

Planned Dissemination Events attendance to present OLEDSOLAR: *(activities to be executed in WP7)*

- Printed Electronics Europe Conference, 2019
- OLEDs World Summit, USA 2020
- PVCELLTECH Event, 2021
- Manufacturing and Technology Conference and Expo, 2021
- Advance Manufacturing, UK
- LOPEC 2019, 2020, 2021 conference and exhibition, Munich
- Swiss e-print 2019/2021 , Basel (+booth)

Exploitation Plan:Table 11: Exploitation plan by partners

Partners	Exploitable Results/Interest	Routes to Exploitation	Potential Uses/Applications	Potential Users and Sectors
Companies				
Smit Thermal Solutions	SALD and PESALD equipment	Product development	Manufacturing of thin films and opto-electronic components	Lighting application, Energy devices sector, optical devices etc.
IRIS	Inspection System	Services/Product	Manufacturing sector	Manufacturing sector
Advent	Recycling system	Service creation	Production sector	Flexible electronics production sector
Coatema	Handling,printing and coating system	Product	Coatinga nd printing based manufacturing	Semiconductors,opto-electronics, etc.
Academic and Research Organisations				
TNO	SALD process	Services/patent	Manufacturing flexible and printed electronics	-Opto-electronics components and devices manufacturing sector, -Semi-Conductors Manufacturing sector -Flexible and printed electronics manufacturing sector
VTT	Printing processes	Services/patent	Manufacturing optical devices	
LETAT	Recycling/Re-use processes	Services/patent	Electronics/glass/metal production recycling application	
TWI	NDT systems	Services	Manufacturing and production of devices	
CSEM	Encapsulation techniques/WVTR measurement technique	Services/ publications	Optical devices manufacturing	
BUL	Data processing, AI and machine learning, software	Services/ publications	Optical devices manufacturing and production	
End users				
ASPF	Advanced OPV manufacturing, inline quality control and machine learning/big data	Implementation of novel processes and tools in OPV production landscape with a direct impact on production output and yield and hence cost reduction	Energy generation, building certification (LEED)	Building integrated photovoltaics (BIPV), Automotive and Transportation, Consumer Electronics, Wearables
SOLIBRO	CIGs manufacturing	Product	*Energy*	Enhanced product development with high yield, efficiency and uptime.
INURU	OLEDs manufacturing	Product	*Displays*	Ehanced products and yields.
DTF	Film manufacturing	Product	*Opto-electronic devices*	Enhanced quality product

Business Plan of Industrial commercial partners:

Partner: Smit Thermal Solutions(SMIT TS)
Planned key exploitable results and key areas of application: Planned exploitable results are spatial atomic layer deposition equipment, plasma enhanced atomic layer deposition equipment, inkjet printing equipment and test benches for pilot lines. In following areas of application: (1) Semiconductor & opto-Electronics, (2) Barrier Layers (3) Integrated Circuit (IC) Applications (4) Solar cells (5) OLED Display Panels (6) Sensors and Research & Development Facilities.
Exploitation Team: Sales Team and Business Development group of SMIT TS.

Targeted Market: The global atomic layer deposition (ALD) market is expected to reach USD 3.01 billion by 2025, according to a new report by Grand View Research, Inc.[46]
Business Model: Direct sales, Equipment renting, sales through distributors, sales through agents globally,
Financial Projection: > 5 million worth of sales in first year of commercialisation.
Commercialisation Roadmap: scale up systems at TRL9 and start commercialisation strategy to enter market in 3 years post project.

Partner: IRIS
Planned key exploitable results and key areas of application: Planned exploitable results are inline optical inspection system and automation kit for factories, production companies, manufacturers of semiconductor, flexible electronics, organic devices, solar cells industry, sensors, MEMs industry, OLEDs industry etc.
Exploitation Team: Sales Team of IRIS.
Targeted Market: The automated optical inspection (AOI) system market was valued at USD 346.4 Million in 2015 and is expected to reach USD 1,008.8 Million by 2022, growing at a CAGR of 17.1% between 2016 and 2022[47].
Business Model: Direct sales B2B and sales through distributors and agents globally,
Financial Projection: > 2 million worth of sales in 2 years of commercialisation.
Commercialisation Roadmap: Commercial scale up and industrial operational system as per industrial specification to TRL9 using our internal resources, investment and product development before commercialisation.

Partner: Coatema
Planned key exploitable results and key areas of application: Planned exploitable results are material handling system, printing and coating equipment for manufacturing of semiconductor, flexible electronics, organic devices, solar cells industry, sensors, MEMs industry, OLEDs industry etc.
Exploitation Team: Sales Team of Coatema and worldwide branches and agents.
Targeted Market: The material handling equipment market worth $48.31 Billion by 2023[48].The Printing Equipment Market forecasted to be worth $22.7 billion by 2017[49].The Coating Equipment Market worth 19.33 Billion USD by 2021[50].
Business Model: Direct sales B2B and sales through distributors and agents globally,
Financial Projection: > 4 million worth of sales in 2 years of commercialisation.
Commercialisation Roadmap: Commercial scale up and industrial operational system as per industrial specification to TRL9 using our internal resources, investment and product development before commercialisation.

Business Model of OLEDSOLAR:

Figure below shows business model(based on Board of Innovation business model) of OLEDSOLAR system. Exploitation of developed components/systems in OLEDSOLAR project will be commercially exploited by SMEs and large companies. End users ASPF, SOLIBRO and INURU will be potential customers of the system and research and academic partners will be technology developer of OLEDSOLAR.The complete set of systems and equipment including software and hardware of OLEDSOLAR will be commercially scaled up to TRL 9 after the project by Smit Thermal Solutions in collaboration with other partners and will be finally commercialised for market uptake. IRIS will commercialise inspection system. Other partners will be supply chain partners for OLEDSOLAR complete system for Smit Thermal Solutions and IRIS as well as they will commercialise standalone unit for various applications.

[46] https://www.grandviewresearch.com/press-release/global-atomic-layer-deposition-market
[47] https://www.marketsandmarkets.com/Market-Reports/automated-optical-inspection-system-market-179056156.html
[48] www.marketsandmarkets.com/Automated/MatrialHandling
[49] https://www.smitherspira.com/news/.../change-in-printing-equipment-leads-to-growth
[50] https://www.marketsandmarkets.com/PressReleases/coating-equipment.asp

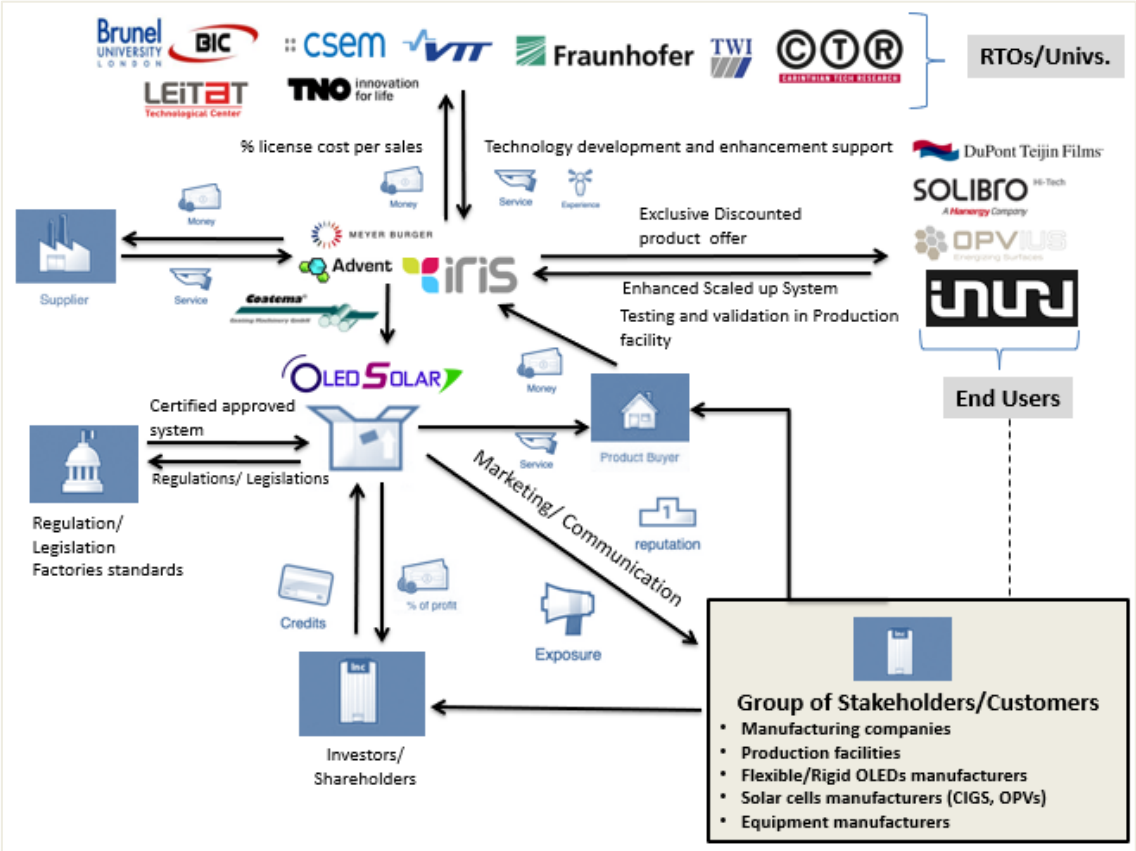


Figure 53 OLEDSOLAR Business Model

Knowledge and data management: OLEDSOLAR project will follow a proactive strategy for internal data and knowledge management. A data repository will be managed by the Coordinator to store common data, results, reports and deliverables. It is expected that OLEDSOLAR will generate some data mainly coming from industrial trials results, but also experimental activities related. Raw data is typically generated by the instruments is standard ASCII format or proprietary format, in former case can be parsed to standards (like ASCII or XML) for easy sharing and cross-platforms compatibility. Confidential data related with the industrial process and products will be kept restricted to the Consortium partners and not published. Not sensitive publishable data will be possibly shared and made accessible to the research community for verification and re-use (e.g. for publications) and cross-validation scientific purposes. Conferences, congresses, workshops exhibitions, and meetings will be included in OLEDSOLAR calendar of events. Based on this, articles will be submitted for publication as soon as some initial results are ready. Partners will preferably address journal articles as the dominant type of peer-reviewed scientific publication for their best results. Especially high impact factor journals. In addition, however, the beneficiaries will encourage "open access" publications for all the other types of scientific publications (some of which may, in some cases, not be peer-reviewed) including monographs, books, conference proceedings and grey literature (informally published written material not controlled by scientific publishers, e.g. reports). In order to comply with "open access" requirement, the beneficiaries will ensure that their publications, if any, can be read online, downloaded and printed. This will be implemented depositing publications into a repository (online archive) and providing open access to it. The right repository will be selected through the Open Access Infrastructure for Research in Europe (OpenAIRE) service or other useful listings such as Registry of Open Access Repositories (ROAR, http://roar.eprints.org/) or the Directory of Open Access Repositories (OpenDOAR, http://www.opendoar.org/).

Intellectual property(IP) Management:

Intellectual property (IP) generated within the project will be protected by patents, if appropriate, and its management will be regulated in compliance with the final consortium agreement and the intellectual property agreement among the partners.

The consortium has identified three levels of Foreground (Results) IP which will be created in the course of the project:

- Individual and joint IP, which belongs to individual partners or is jointly owned by partners working in a particular task and is restricted to those partners.
- Generic IP, which can be used by all partners of the consortium. Some of the generic IP will be made more widely available to European academics, SME's and other industrial organizations.

- Publicly available IP which will be published at conferences, on the public web site and made available with no restrictions.

To prepare for the deployment and exploitation of the project results, the IPR/Exploitation leader will constantly monitor the progress of the research, as well as the advances outside the Consortium with support of the Executive Board, for an early monitoring of resulting knowledge which should be protected, and for a continuous updating of the research/work programme to improve project effectiveness.

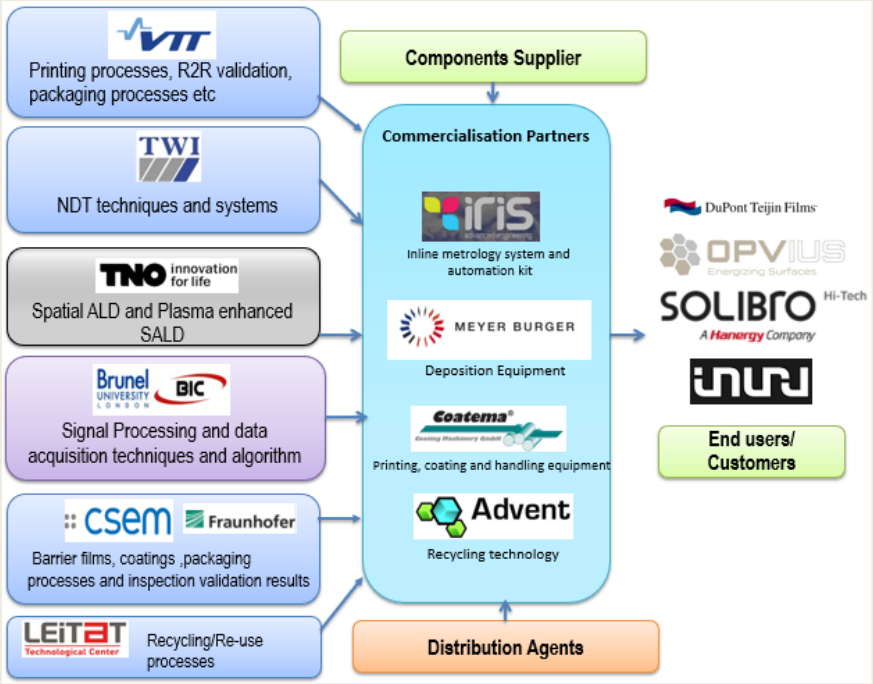


Figure 54 Foreground IPs and commercialisation partners

The IPR/Exploitation leader Smit Thermal Solutions with support of the Executive Board (comprising of one members of each partner in consortium) will be responsible for the delivery of the following tasks:

- Creating and maintaining a register of background IP belonging to the consortium members and made available for the purposes of the project.
- Creating and maintaining a register of foreground IP generated during the project, to include: ownership, restrictions on access, licensing arrangements.
- Working with the consortium partners to extract generic knowledge from foreground IP to enable access by all consortium partners for exploitation and for dissemination and training purposes.
- Moderating disputes arising between consortium partners on IP and knowledge management related issues.
- Reporting to and advising the management team on issues related to IP and knowledge management.

These activities will ensure a transparent and fair handling of IPR issues.

The general principles for Intellectual Property Aspects set out by the EC for Horizon2020 projects will apply in OLEDSOLAR. Existing know how (background or pre-existing intellectual property) of a specific partner shall be made available on transfer conditions to the Partner(s) within the consortium. The use of such existing know-how is strictly limited for use to the achievement of the project goals and for the duration of the project. An overview of the existing know how will be included as an annex to the consortium agreement. Foreground shall be owned by the partner(s) who developed these results. Each partner is responsible for taking the appropriate steps for securing intellectual property of the knowledge or results created during the project.

Results: As a direct consequence from the project owned by one or more of the partners shall be licensed to other partners of the consortium on favourable conditions to the extent necessary to enable these partners to exploit their own results.

Table 12: Foregorunds of project

Description of expected Result and/or IPR Outcome	Nature or Type
Three pilot lines	Demonstrators
Development of new front-end manufacturing processes: Integration of Atomic Layer Deposition into a production process sequence for complete protection of cells against humidity	Hardware/New techniques

Development and integration of printing processes for OPVs	Hardware/New techniques
Development and integration of inkjet printing process for OLEDs	Hardware/New techniques
Material handling system	Hardware/New techniques
Coating system	Hardware/New techniques
Automation kit	Hardware/New techniques
In-line monitoring system (hybrid optical system)	Methodology and hardware model
Analytics software for production line	Software
Methodology for the automatization of the opto-electrical characterization of solar cells and OLEDs	Methodology/Report

Publication of results:

The partners are entitled to publish the project results in the usual scientific form. However, all concept publications must be put on the website and submitted to all Partners together with a request for permission to publish. Requests for such permission to publish shall be responded to within one month of receipt thereof. An agreement is considered to have been granted if no objection is raised within a period of one month after submission of the manuscript to all partners. One of the pages of OLEDSOLAR website will contain an overview and archive of all published information: scientific articles, publications, press releases, conference papers, etc. Publications will be submitted to reputable journals (using peer-review) that can provide "gold" open access (immediate open access mode) to reach the broadest possible audience for the results within a short space of time. More details on these issues will be described in the Consortium Agreement.

b) Communication activities

Activities to disseminate information and exploit research and innovation results as well as carry out communication activities will be an important and integral part of Horizon 2020. *(European Commission Decision C (2013)8631 of 10 December 2013).* The following communication measures and strategies will be adopted for promoting the project at European level and to reach the largest possible audience, including groups beyond the project's own community. Communicating relevant knowledge about the project's latest activities and achievements to the relevant identified OLEDSOLAR target audiences is certainly a way to keep all partners actively involved in the project and the OLEDSOLAR consortium accordingly realizes that communication is not something that should be dealt as a side task; on the contrary, it is one of the topmost aspects, priorities and prerequisites of running a successful research project. In this context, the draft communication plan for promoting this project and its results, include an efficient and effective mix of both interpersonal and mass communication tools. Moreover, it shall be strongly emphasized, that for each targeted audience, a distinct strategy using targeted messages, means and language has been planned. Namely, specific audiences and communication methods have been defined for promoting OLEDSOLAR and its main results:

- **Direct proactive communications and physical demonstration:** *Attendance in seminars and conferences, one-to-one communication, e-mailing stakeholders, periodic newsletters, etc.;*
- **Clustering activities:** *Contacting parallel related projects, cross-field events;*
- **Mass and general communication:** *Project website, press releases, posters and leaflets;*
- **Scientific excellence:** *Publications in highly-ranked journals, presentations in conferences;*
- **Academic:** *Conveyance of the new knowledge into University curriculums, publications of MSc and PhD thesis.*

Below, some of the primary communication activities to be performed during the period of the grant are described, as well as the main target audiences to whom they will be addressed to:

Electronic and printed dissemination material: Appealing and accessible dissemination materials such as logos, posters, banners, leaflets and newsletters will be developed by coordinator VTT to structure the "visual identity" of OLEDSOLAR. Printable material with information for different target groups will be provided to all partners to distribute at conference venues, while electronic copies will be used for online publications, articles and other references such as local and international press. Additionally, a dissemination pack for the project legacy will be created and used by all partners as an advertising tool for the work performed by each partner within the project in events performed after the end of the project.

Project website: A dedicated OLEDSOLAR **public website** will be developed for providing information about the project's objectives, progress and results for the different target groups (from consortium members – general public). Care will be taken to present information in a comprehensible and attractive manner, thus allowing grand access to citizens. The project website will also serve as a gateway to a private (password protected) collaboration platform for

the consortium partners. The website will be continuously updated during the project lifetime and will be maintained for a minimum of one additional year after the project ends. VTT will provide connection between the project website and will assist Project consortium in linking their web pages with the project and with other portals as CORDIS WIRE, Horizon the EU Research and Innovation Magazine, etc.

Regular Press Releases, Newsletters and Other Publications: All partners will be responsible for publishing project results in local and international press. These **publications** could be in the form of papers in scientific journals and conferences, press releases or newsletters in magazines and newspapers, etc. It is worth mentioning that special emphasis will be placed on publishing project-related information in **Open Access journals** which provide to the readers on-line access to scientific information free of charge. VTT will maintain an overview of all published results and in cooperation with the project manager will intervene in case of results which have been classified as publishable do not receive the necessary dissemination.

Organization and participation at workshops and relevant events: The presence of the project, its main objectives, major results and offered services; shall be shown in numerous suitable events for dissemination. In addition, understanding the importance of face-to-face communication and physical trials, VTT will arrange at least one major International Workshop of relevance. Further to that, and if project timing permits, a joint broader International Manufacturing Conference shall also be organized, in order to exploit the already mass dissemination and outreach for amplifying the consortium's performed work.

Clustering activities: Contacting related and parallel projects will contribute in dissemination and exploitation opportunities, while at the same time support exchanging technology novelty and ensure maximum complementarities. Such clustering activities may be linked to the above point regarding workshops and conferences, leading to the organization of a conference for maximizing synergies.

OLEDSOLAR Target Groups/End users: As already stressed above, an extremely imperative precondition so as to ensure augmented exploitation, high impact and increased likelihood of uptake of the project's results, is to prudently and effectually disseminate and communicate the appropriate information to the relevant and interested audience; in a concise, well-articulated, understandable and attractively packaged manner. Hence communicating the right information and messages to the right people, using the right language and suitable means **is the key to a successful dissemination strategy**. In this context, by clearly identifying and classifying the different OLEDSOLAR target groups, the messages and information to be consigned to each of them as well as the communication language, measures and channels with which they shall be sent to each tier of audience, **the likelihood of uptake of the project's findings is exceedingly increased.**

In this regard, the following table presents the main dissemination and communication tools to be used for effectively diffusing project related knowledge and information to each OLEDSOLAR target audience/end user:

Table 13: Communication activities

Communication Measures/Actions	Target Audience	Timescale
Project Website,	Research, Scientific & Academic Community, Commercial & Industry, Investors, General Public, Policy Makers	Month 3
Conference, Seminar	Research, Scientific & Academic Community, Commercial & Industry, and Policy Makers	Periodically during the Project Timeline
Wikipedia/Cordis/EU Webpages	General Public and Research, Scientific & Academic Community	Month 3
Project Newsletter	Research, Scientific & Academic Community, Commercial & Industry, Investors and Policy Makers	Newsletter Annual Edition in Months 12, 24, 36,
Clustering Activities	Research, Scientific & Academic Community, Commercial & Industry, Investors and Policy Makers	Month 12,24,36
Workshops	Research, Scientific & Academic Community, Commercial & Industry, SMEs	Relevant Workshops organised during project timeline
Media, Press Releases	Investors and General Public	Periodically during the Project Timeline

Poster, Leaflets, Presentations, Dissemination Support Material	Research, Scientific & Academic Community, Commercial & Industry, Investors, General Public, Policy Makers	Month 3

3 Implementation

3.1 Work plan — Work packages, deliverables

Brief presentation of the overall structure of the work plan:

The brief overview of workplan and interconnection between different WPs is shown in PERT figure below. The project starts with WP1 establishing requirements of high yield manufacturing parameters, inspection needs and material challenges to achieve reliability of components and longer lifetimes promoting sustainability thus less waste and material to be recycled thus contributing to circular economy. Further, the sensors and automation of R2R and S2S lines will also be analysed in WP1. In WP2, innovative deposition, packaging and patterning techniques and process will be optimised for industrial scaled up implementation.WP3 will deal with inspection system, techniques and software development.WP4 will incorporate automation, sensor system, hardware and intelligent decision management software. WP5 will develop recycling and re-use strategies and technologies to promote sustainability. WP6 will include testing and trial at pilot scale as well as integration and validation activities in production lines of end user partners. WP7 will include dissemination and exploitation activities and WP8 will outline project management, coordination guidelines and implementation.

Graphical presentation of the components showing how they inter-relate (Pert chart):

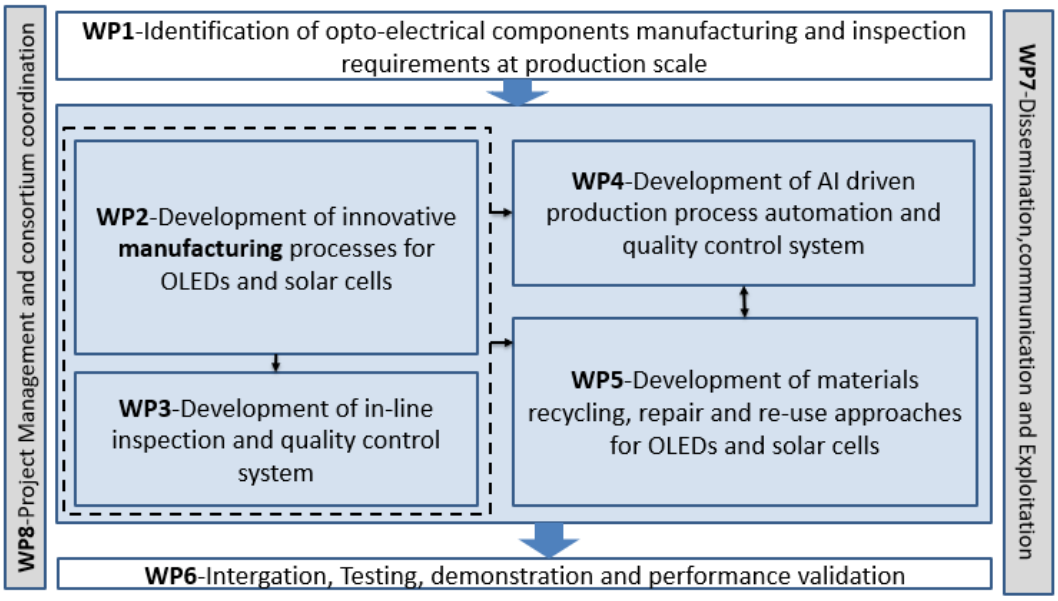


Figure 55 Project PERT diagram